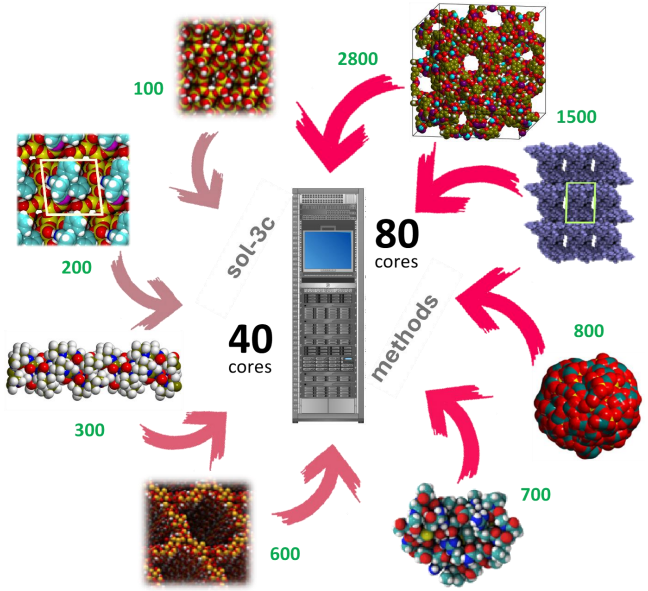
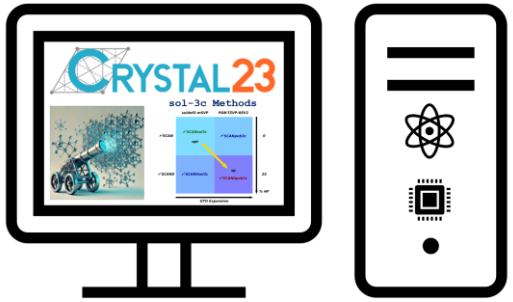
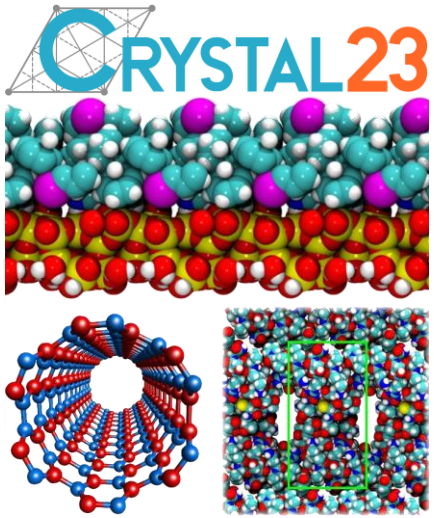
MSSC2025

Cost-effective Composite HF/DFT Methods for High Throughput Screening and Machine Learning Applications

Mattia Raimondo



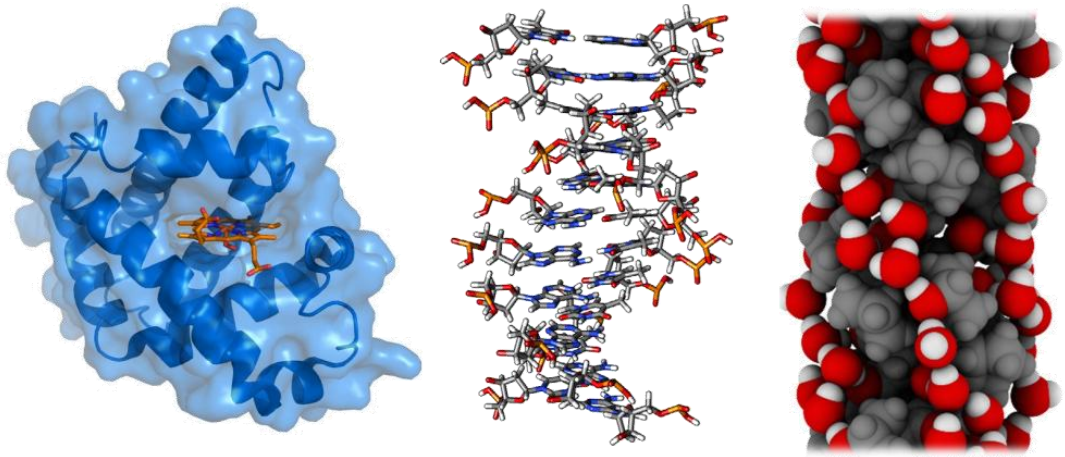




September 8 to 12, 2025
Turin, Italy

Large-scale modelling of materials

Large & Complex systems



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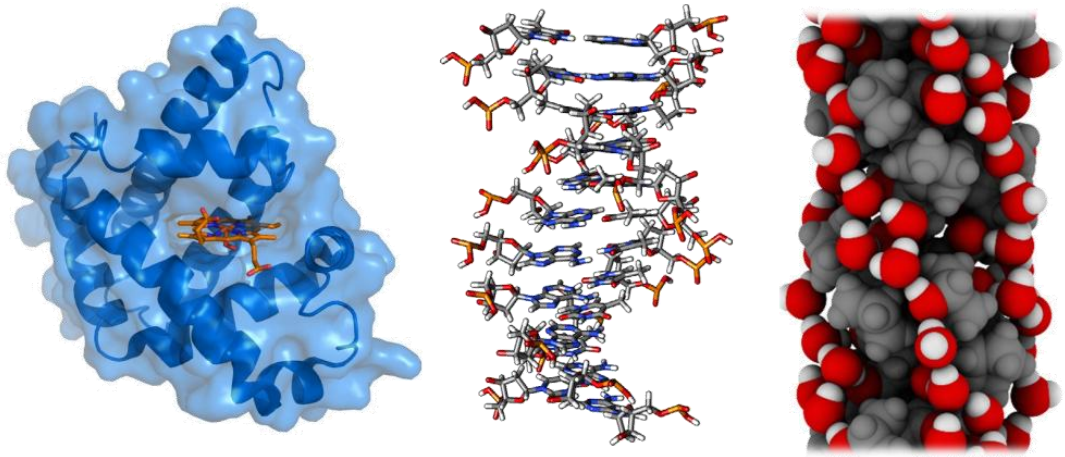
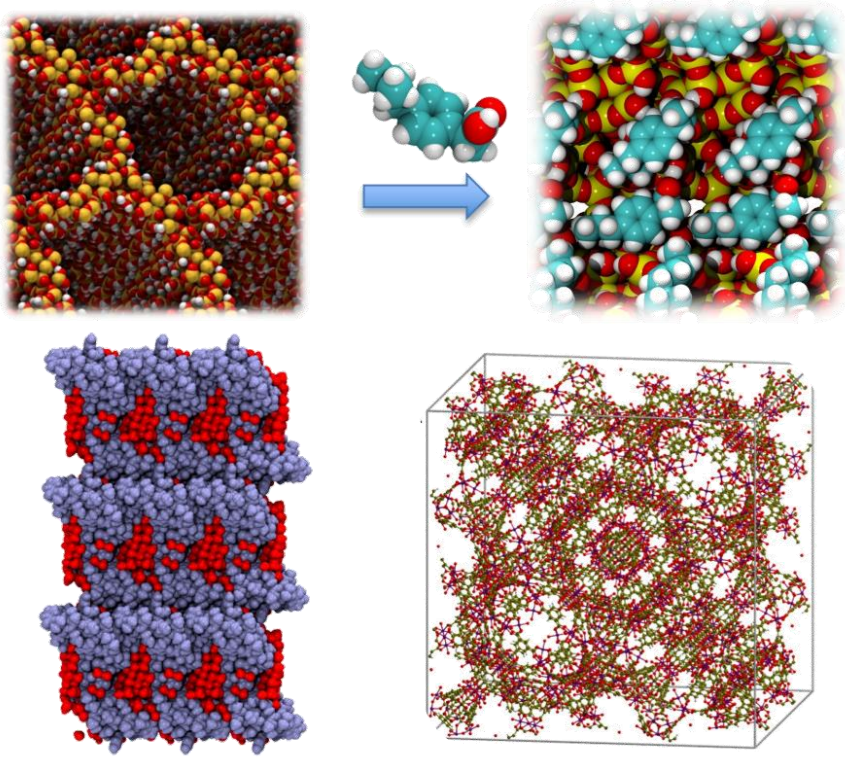


Large-scale screenings:

- ***Drug design***
- ***Crystal Structure Prediction***
- ***High-Throughput Screening***



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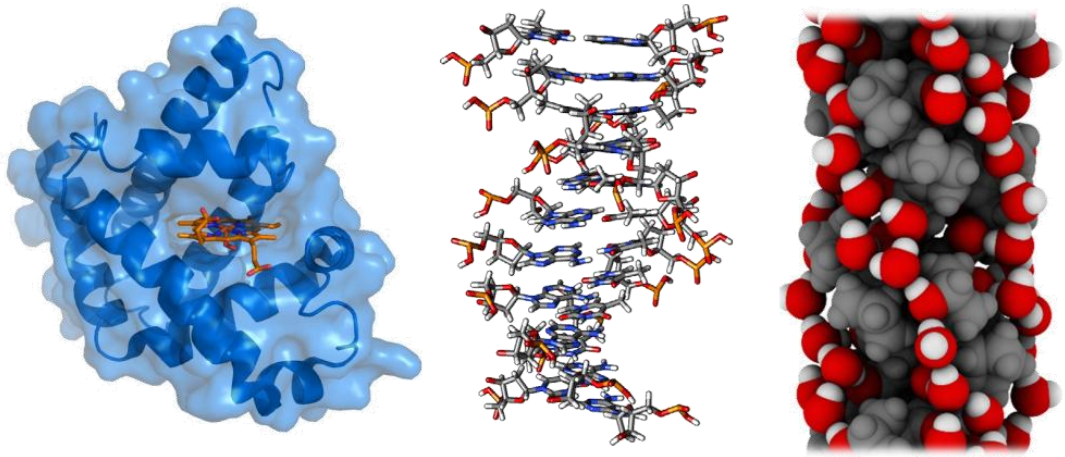

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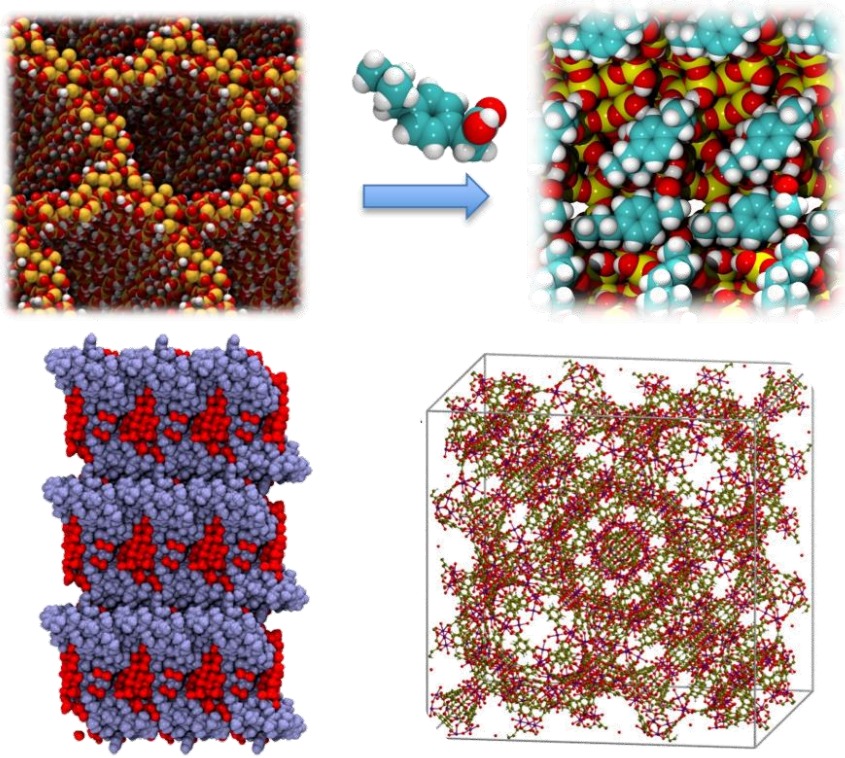

1) Are DFT calculation affordable even with m-GGA and hybrid functional?

2) How far can we go with small-scale computing resources?

Low cost, yet accurate methods are needed

S. Grimme

ab initio composite methods: idea

J. G. Brandenburg

Hierarchy of simplified electronic structure methods for good structures, reasonable energetic properties and non-covalent interactions.

$$E_{tot}^{3c} = E_{tot}^{DFT/basis} + E_{disp}^{D3} + E_{BSSE}^{gCP} + E^{SRB}$$

Key ingredients:

- ***Based on HF and DFT***
- ***From minimal to small-to-medium basis sets (atom centered BF, GTO)***

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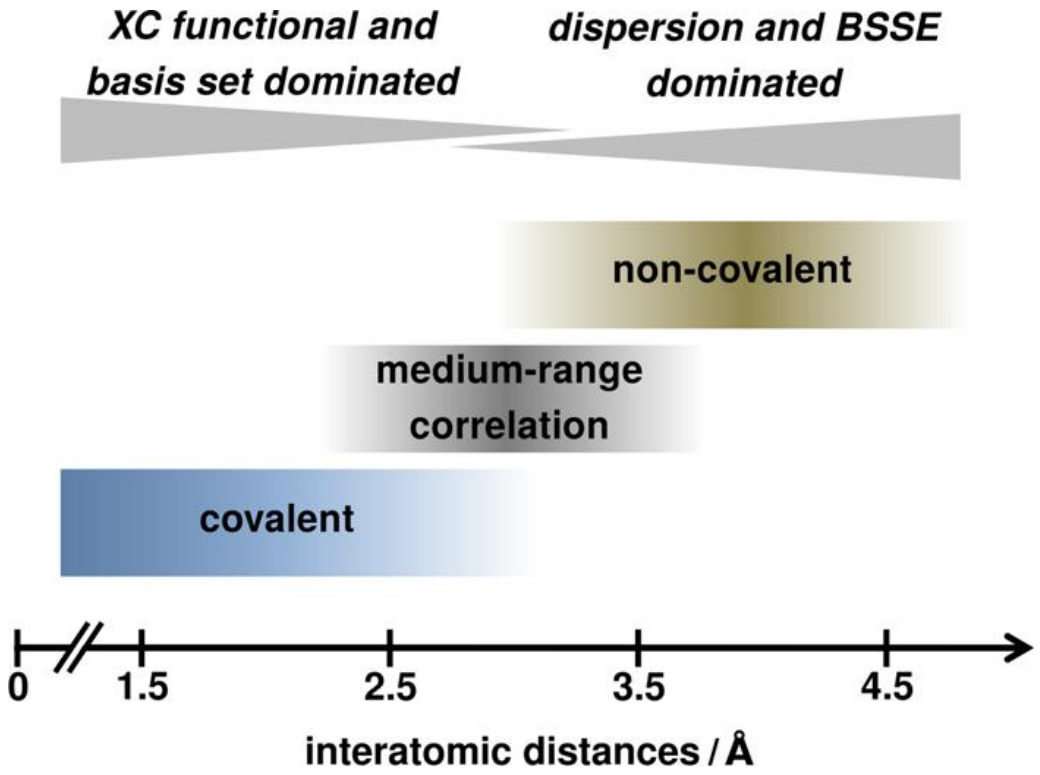

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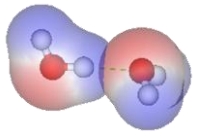

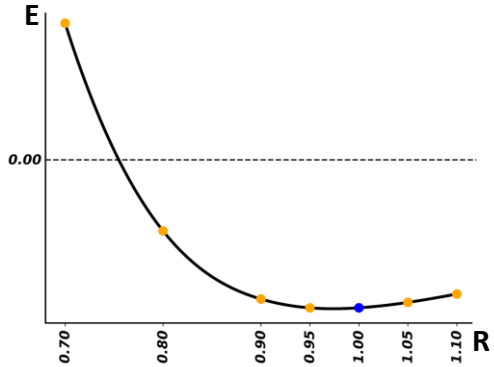

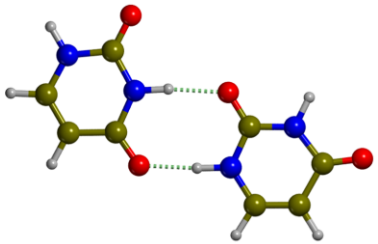

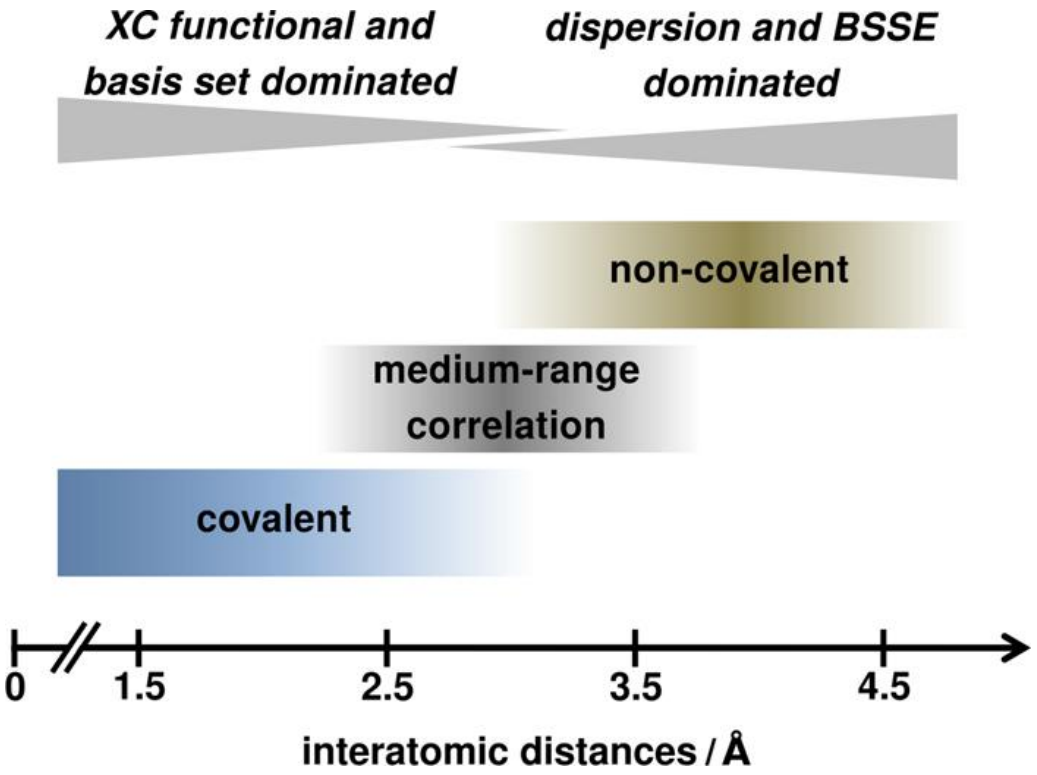

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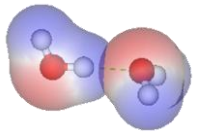

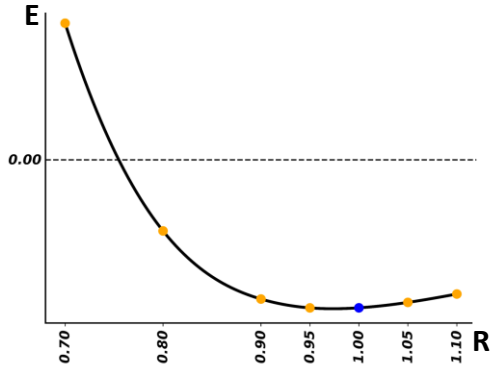

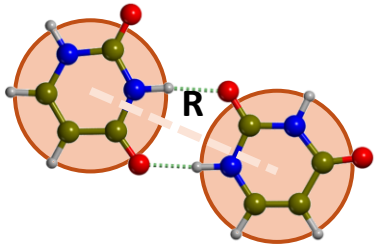

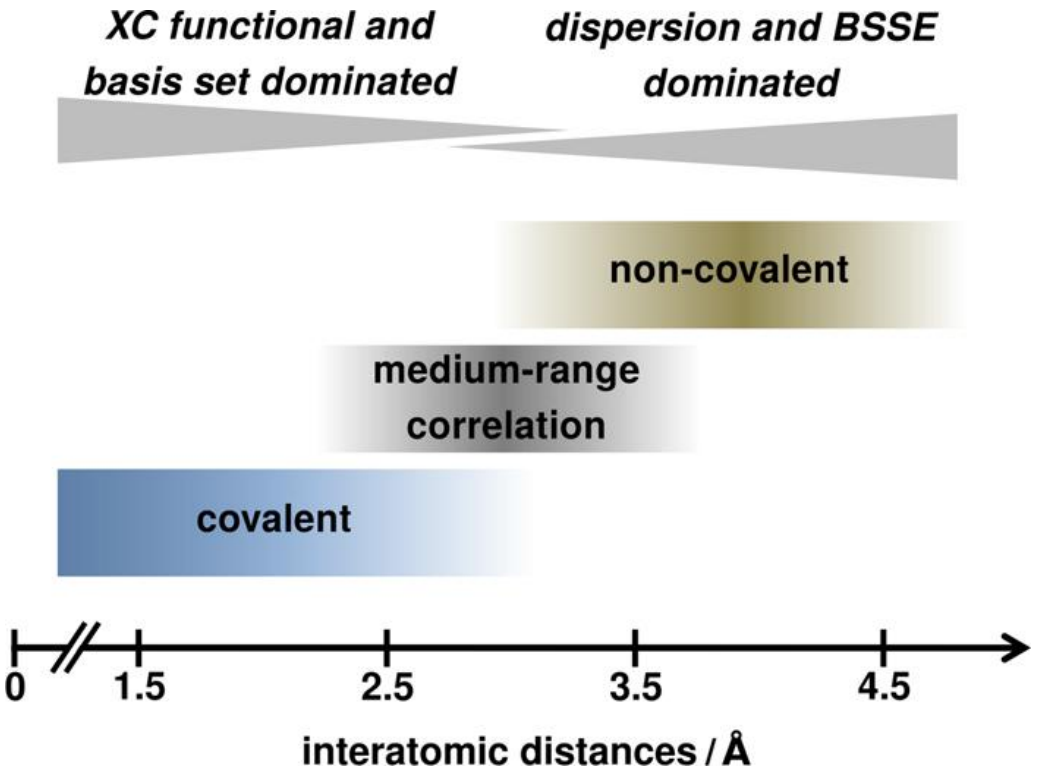

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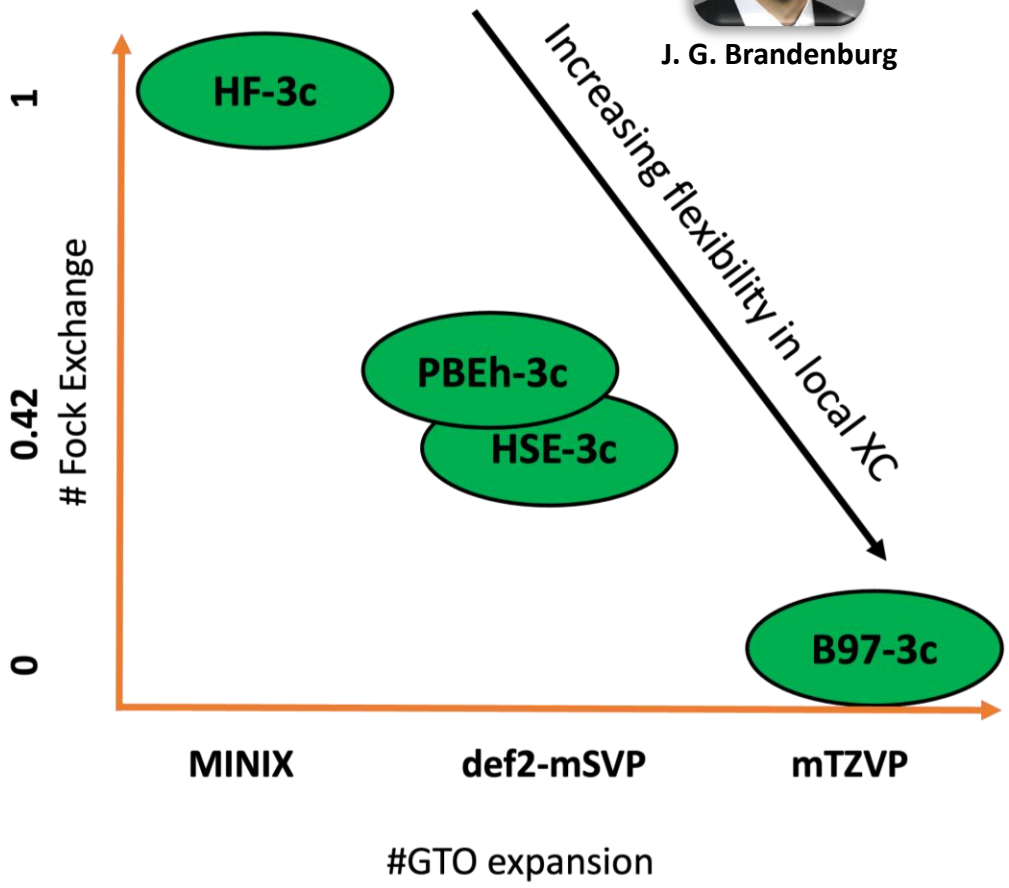
	HF-3c	PBEh-3c	HSE-3c
Hamiltonian	Hartree-Fock	Global hybrid	Range separated
Basis set	MINIX	def2-mSVP	def2-mSVP
Dispersion	D3 (2-body)	D3 (2&3-body)	D3 (2&3-body)
BSSE	gCP	gCP (damped)	gCP (damped)
BSIE	SRB	-	-



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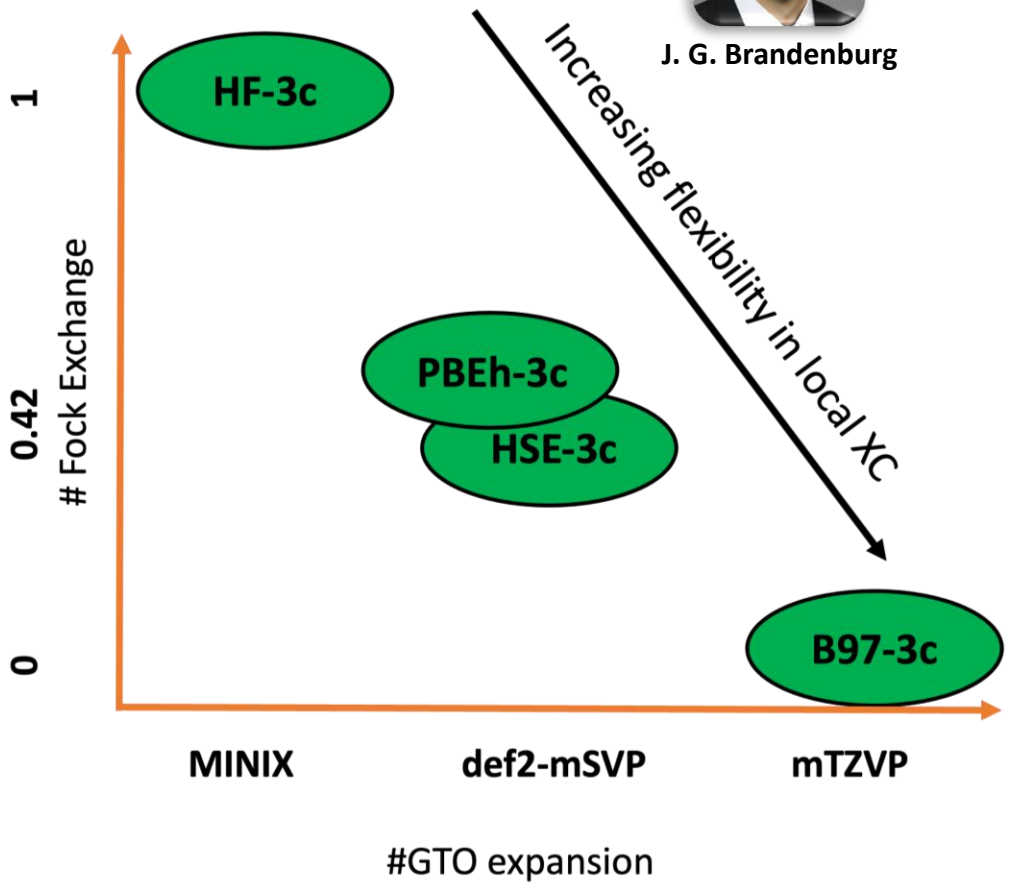


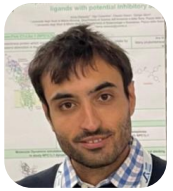
Accurate for:

- ***Protein-ligand binding affinity***
- ***Molecular crystals with short hydrogen contacts***
- ***Light elements of the Periodic Table***

Not applicable to:

- ***Covalent and inorganic solids***
- ***Layered and porous materials***
- ***Heavier elements***

Extending applicability to solid state: sol-3c

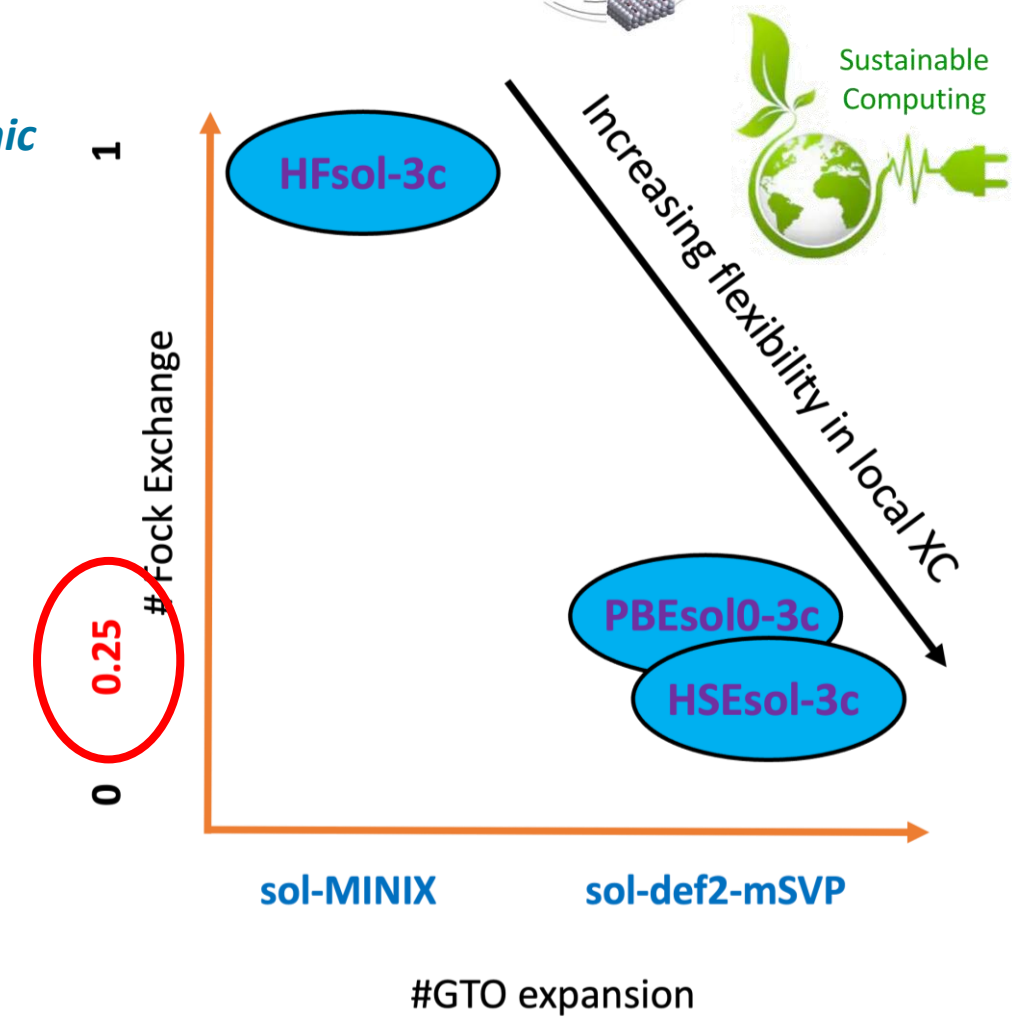


L. Donà

$$E_{tot}^{sol3c} = E_{tot}^{DFT/basis} + E_{disp}^{D3} + E_{BSSE}^{gCP}$$

Methods (hybrids)

- *Employ XC functionals devised for solids (PBEsol, HSEsol)*
- *Reduce the amount of HF exchange for a better description of electronic and optical properties*



L. Donà, B. Civalleri et al., J. Chem. Phys., 151, 121101-121106, **2019**

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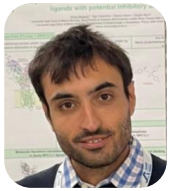

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- *MINIX: Decontract outermost shells and scale most diffuse functions*
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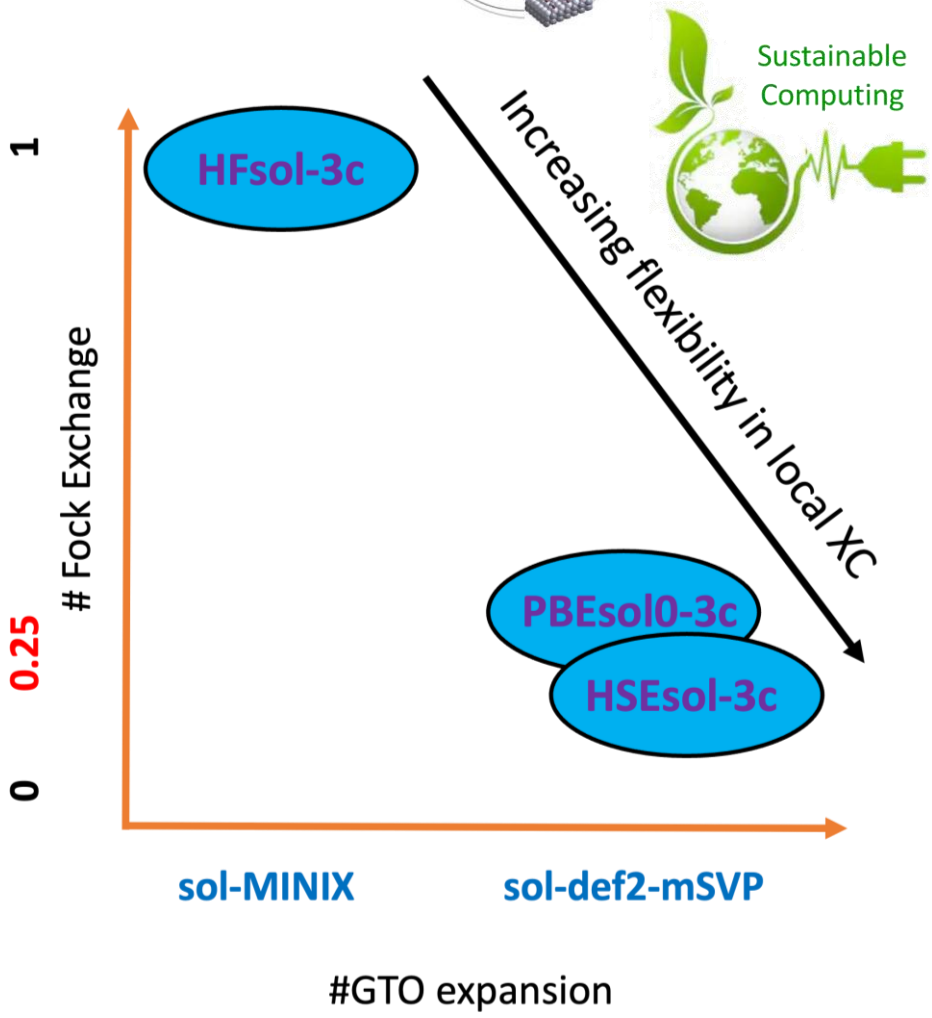

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CRYSTAL23

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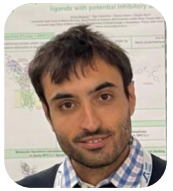

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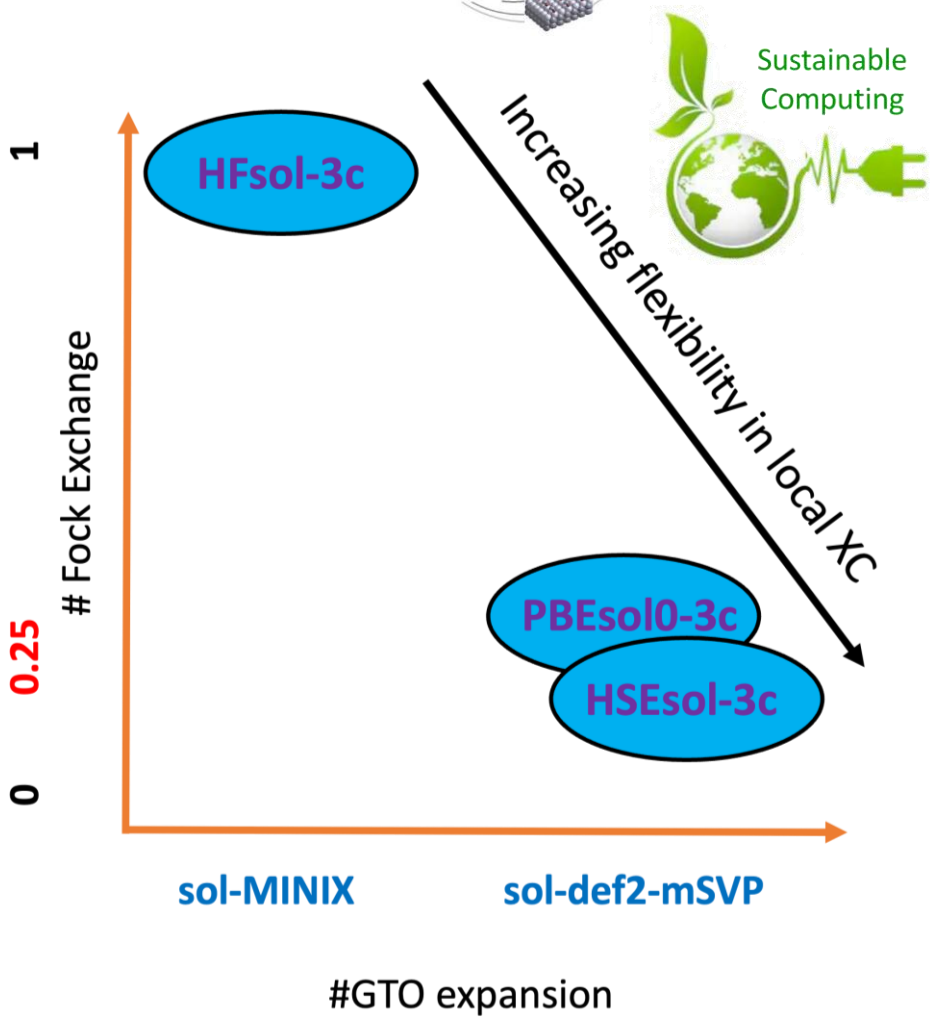

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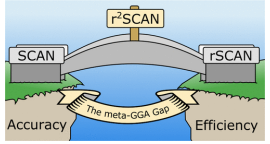
r²SCAN-derived composite methods

$$E_{tot}^{sol3c} = E_{tot}^{DFT/basis} + E_{disp}^{D3} + E_{BSSE}^{gCP}$$

- ***Based on DFT (mGGA & hybrids): r²SCAN/r²SCAN0***



- ***From small-to-medium basis sets (atom centered BF, GTO): soldef2-mSVP & POB-TZVP-rev2***

- ***London dispersion through D3(BJ) correction***
- ***BSSE counterpoise by gCP correction***
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M. Raimondo, L. Donà, B. Civalleri, "r²SCAN-based composite methods for solid state calculations" in preparation.

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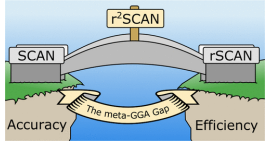

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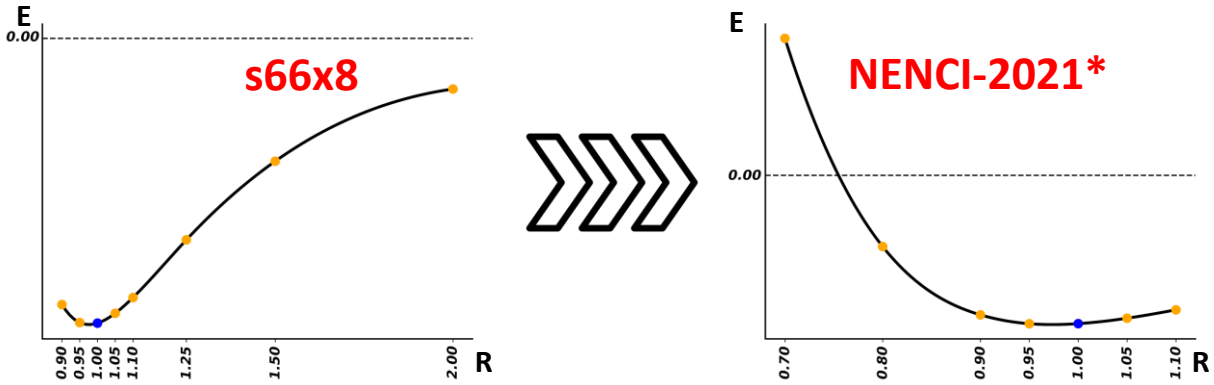

M. Raimondo, L. Donà, B. Civalleri, "r²SCAN-based composite methods for solid state calculations" in preparation.

Z. M. Sparrow et al., J. Chem. Phys., 155, 184303-184321, **2021**

r^2SCAN-derived composite methods

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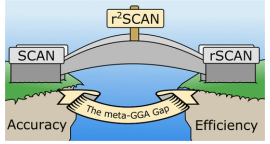
Sustainable Computing

- *Based on DFT (mGGA & hybrids): r^2SCAN/r^2SCAN0*



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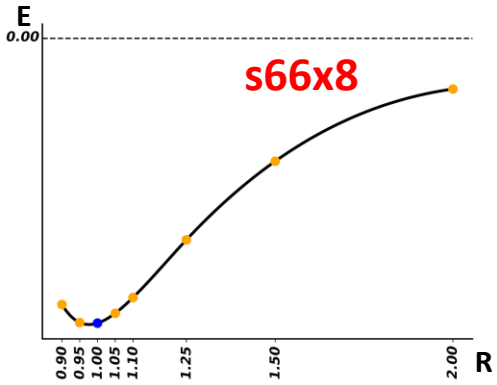

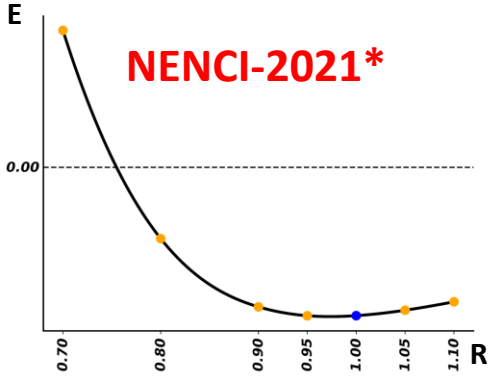

	soldef2-mSVP	POB-TZVP-REV2	
r^2SCAN	r^2SCANsol3c	r^2SCANpob3c	0
r^2SCAN0	r^2SCAN0sol3c	r^2SCAN0pob3c	25
			% HF

GTO Expansion

M. Raimondo, L. Donà, B. Civalleri, "r^2SCAN-based composite methods for solid state calculations" in preparation.

Z. M. Sparrow et al., J. Chem. Phys., 155, 184303-184321, **2021**

sol-3c composite methods: validation

Have these methods the same accuracy of the original ones?

G. A. Dolgonos, J. Hoja, A. D. Boese, Phys.Chem.Chem.Phys., 21, 24333, **2019**

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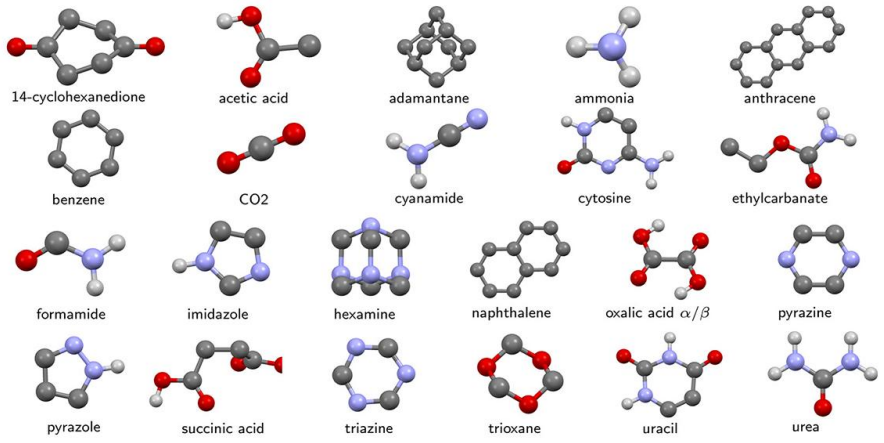

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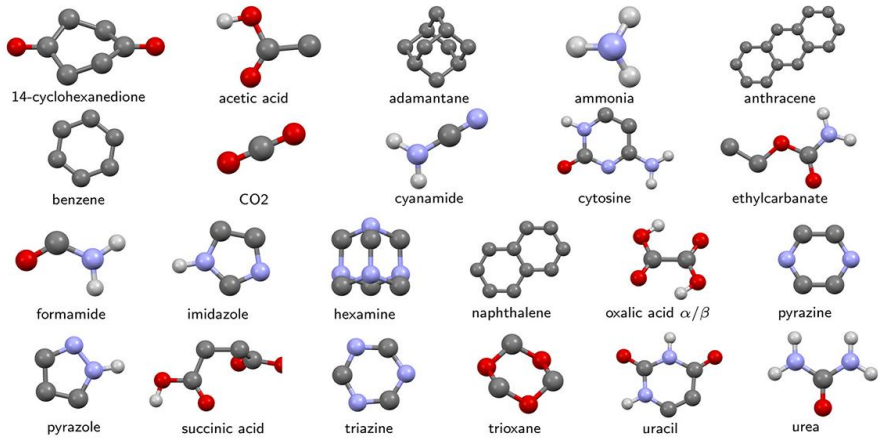
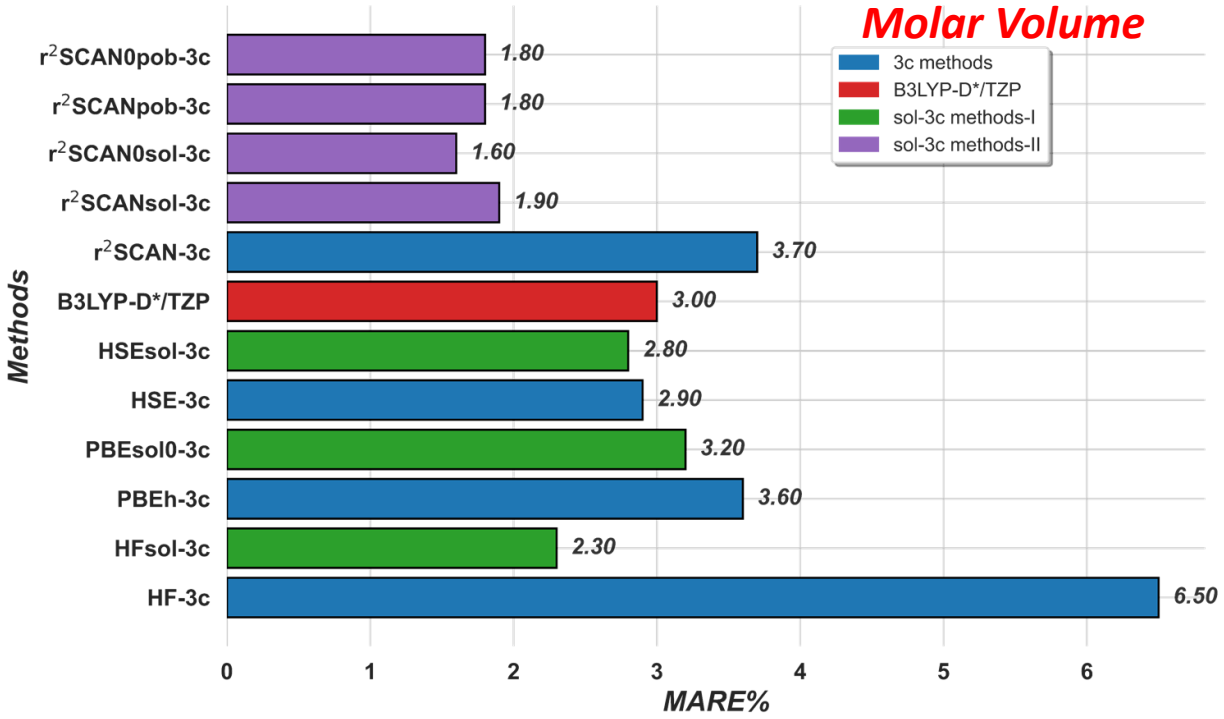
- **23 molecular crystals**: low-T crystal structure (X-ray/Neutron), accurate sublimation enthalpy (298K), no phase transitions (0<T<298K)
- Comparison with back-corrected (ZPVE and thermal contribution from DFT) experimental ΔH (Range: 28 < ΔH < 160 kJ/mol)

G. A. Dolgonos, J. Hoja, A. D. Boese, Phys.Chem.Chem.Phys., 21, 24333, **2019**

sol-3c composite methods: validation





Have these methods the same accuracy of the original ones?

- **23 molecular crystals**: low-T crystal structure (X-ray/Neutron), accurate sublimation enthalpy (298K), no phase transitions (0<T<298K)
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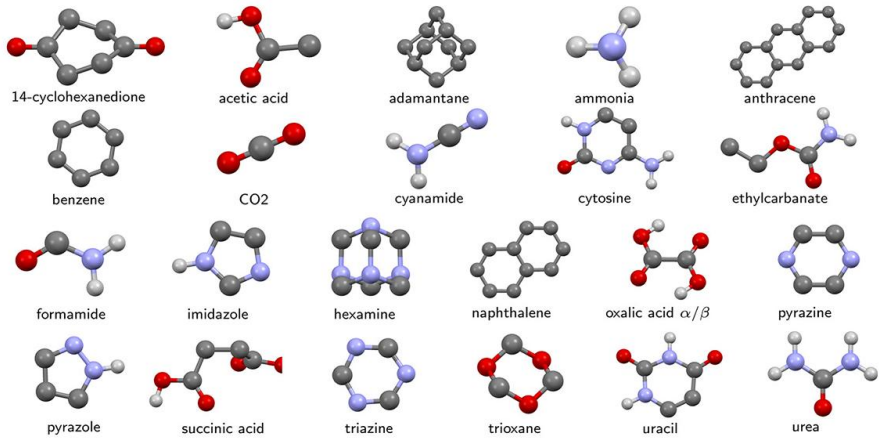

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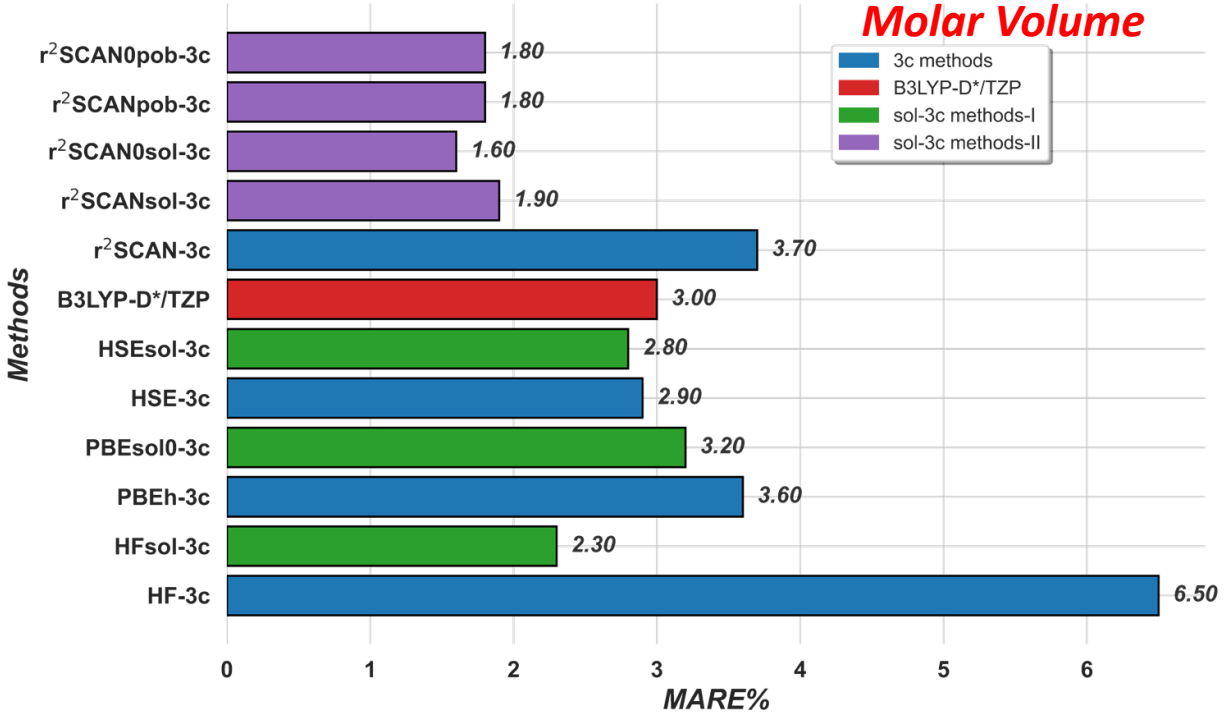
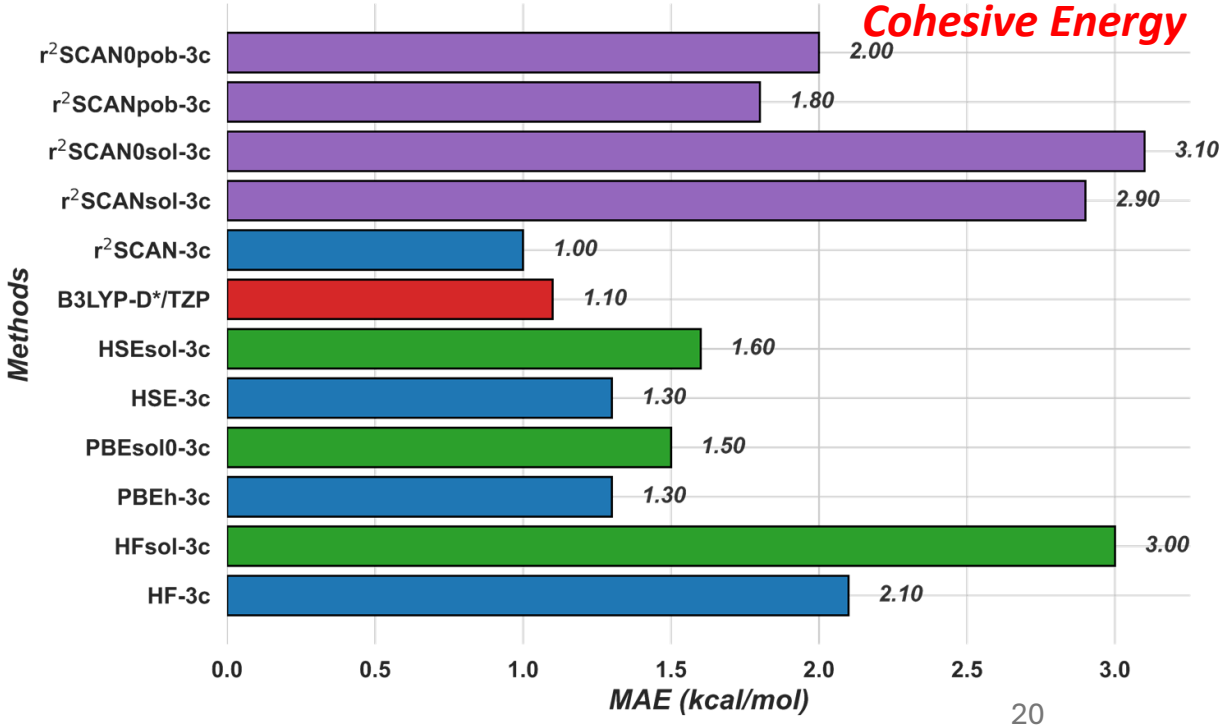
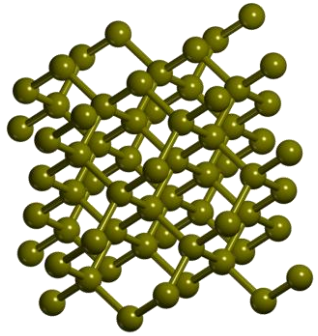




G. A. Dolgonos, J. Hoja, A. D. Boese, Phys.Chem.Chem.Phys., 21, 24333, **2019**



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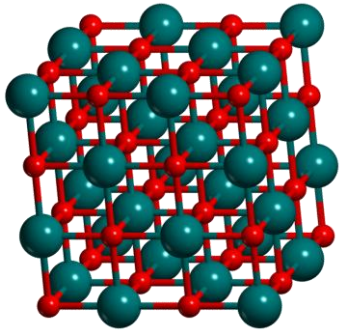
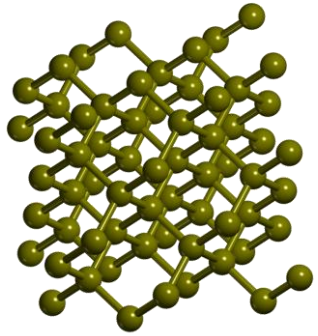


Are these methods of general applicability?



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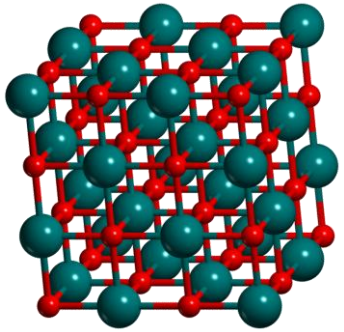

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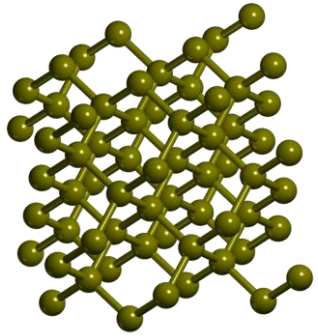
SS20*: 20 simple solids – cubic lattice – covalent and ionic systems – Low-T data

The presence of very diffuse Gaussian functions meant that the original 3c methods were not suitable for inorganic solids

P. Pernot, B. Civalleri, D. Presti, and A. Savin, J. Phys. Chem. A, 119, 21, 5288–5304, **2015**



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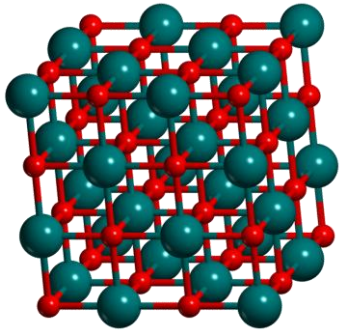

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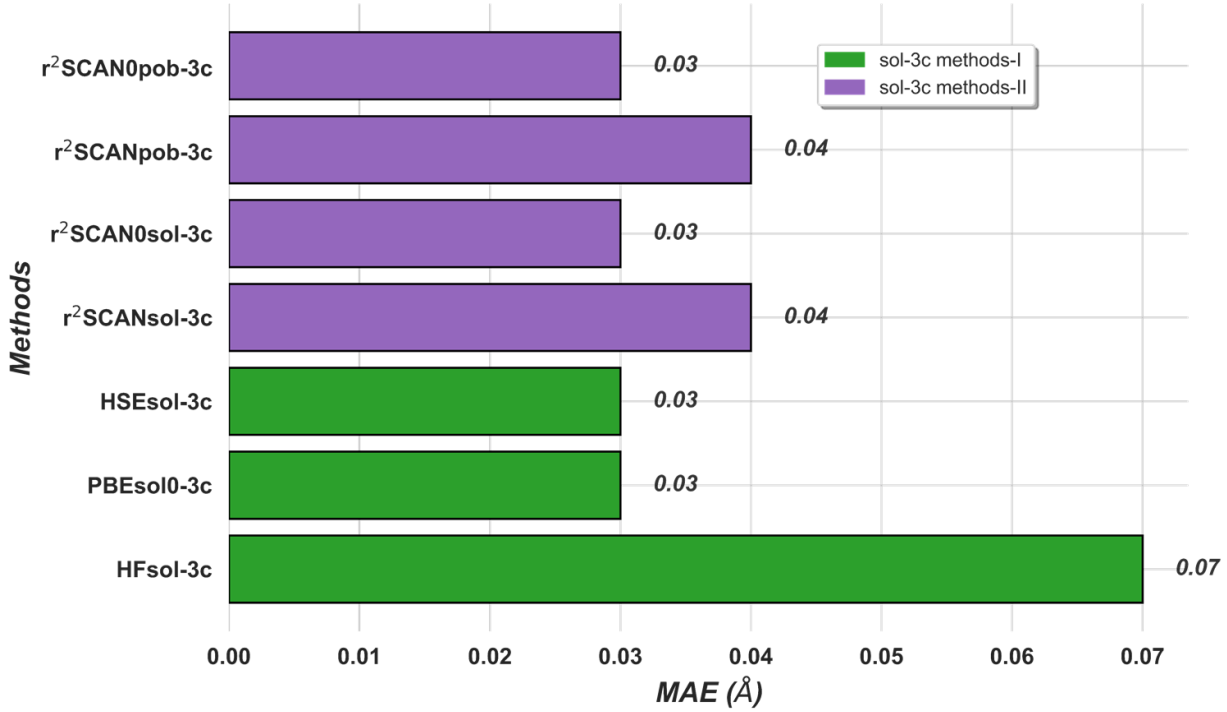
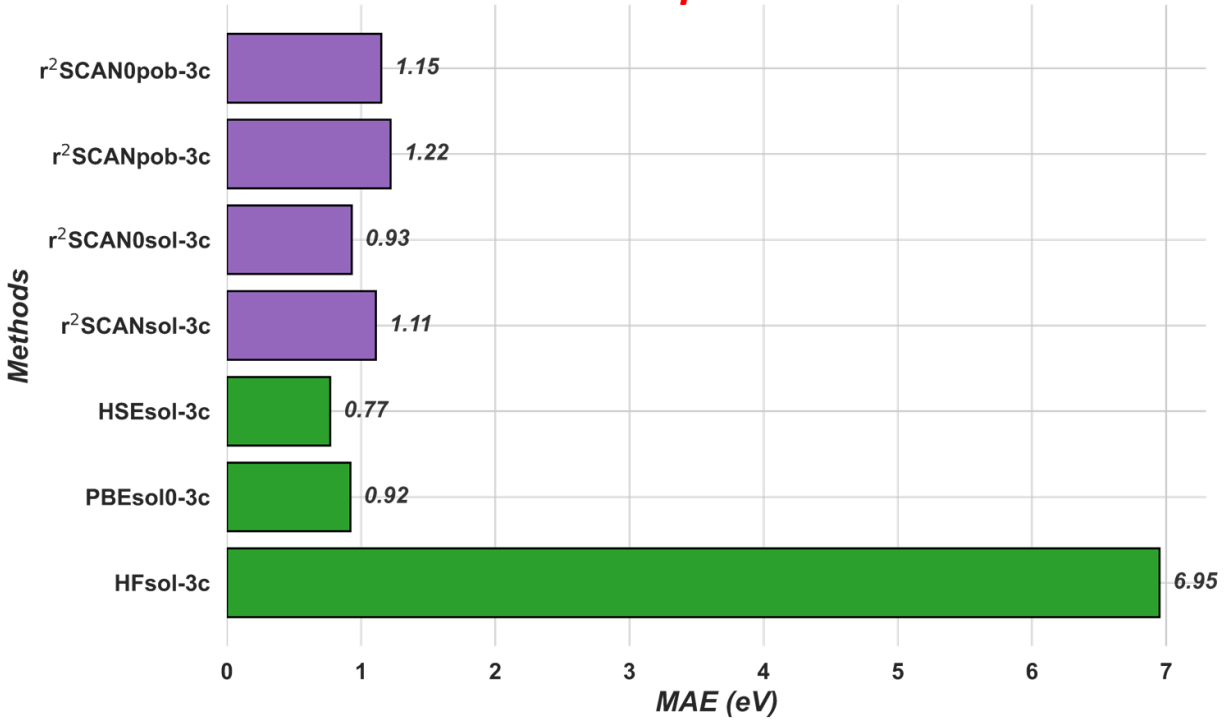
Lattice Parameter



Band Gap



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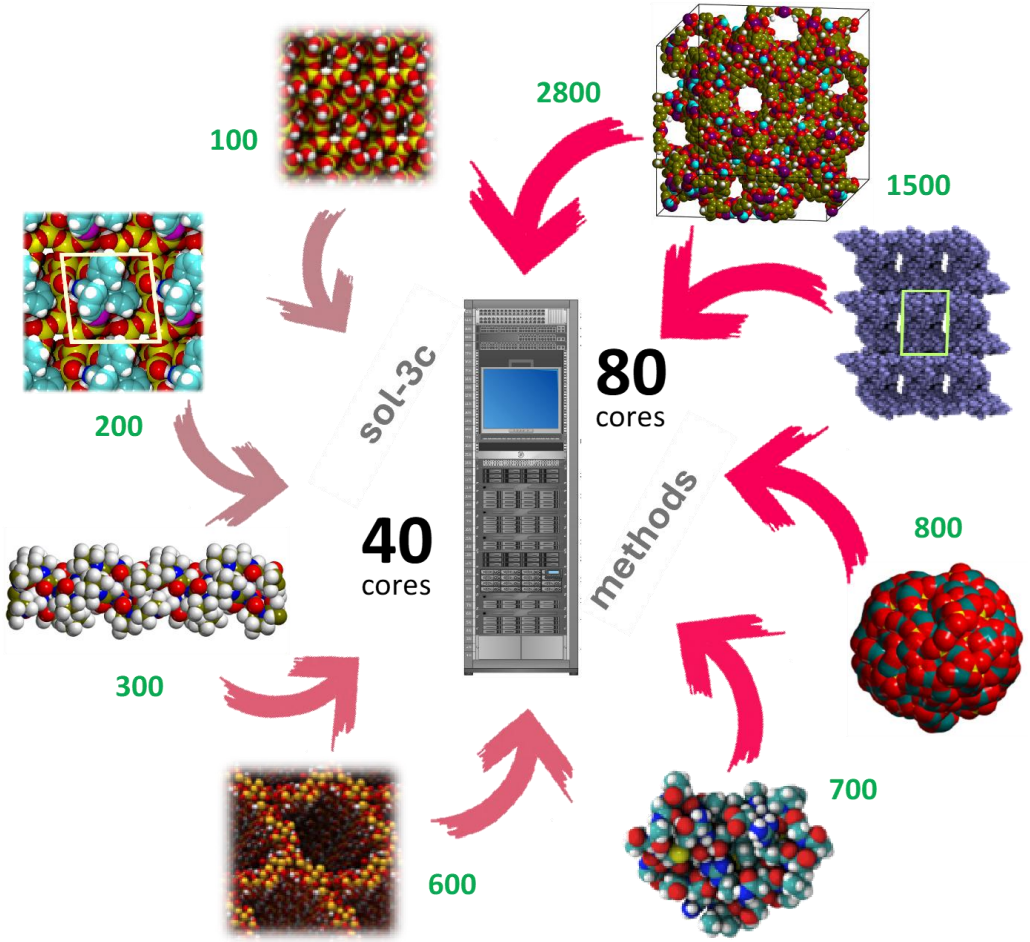
Large Scale Screening

Are these methods really cost effective?

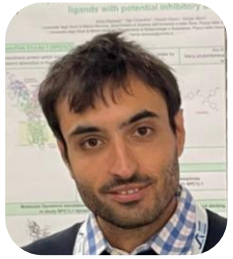


System types:
- 0D to 1D, 2D and 3D;
- molecular crystals and inorganic solids;
- from dense to porous materials

System sizes
- Up to **1.5k atoms** (16 k AOs) on **40 cores**
- Up to ≈**3k atoms** (37 k AOs) on **80 cores**

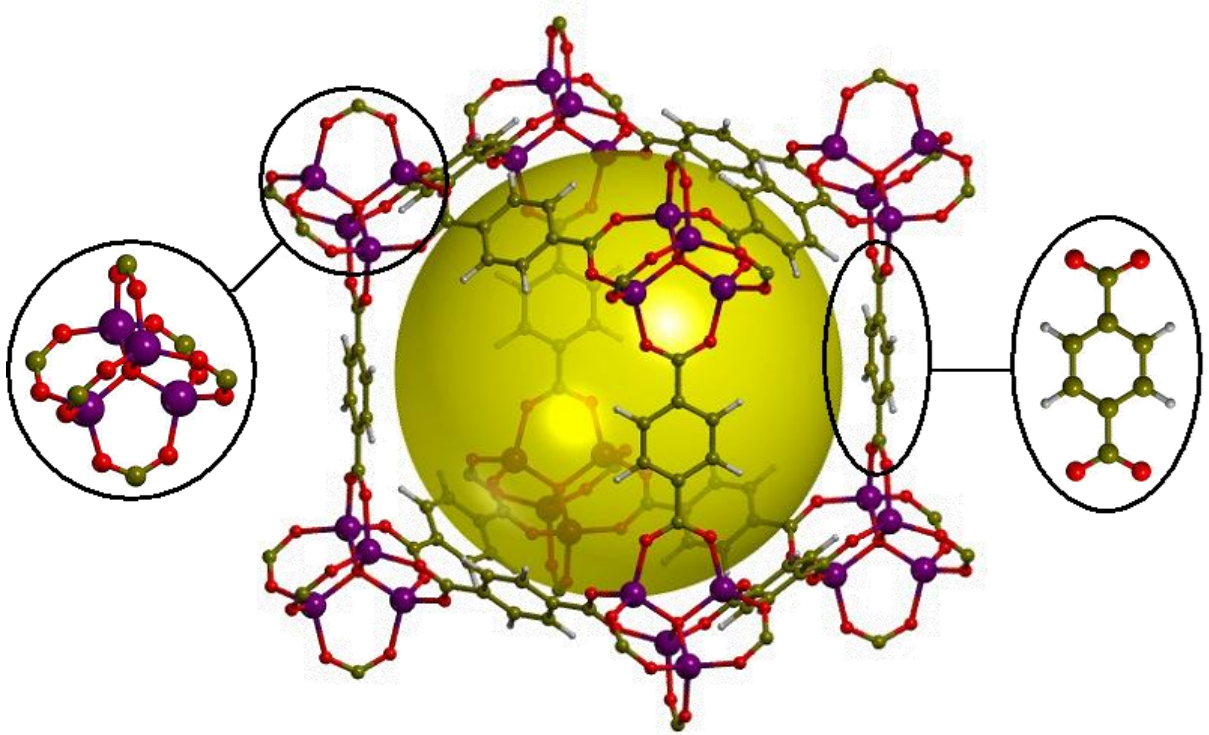
How to Model Metal-Organic Frameworks with CRYSTAL

Lorenzo Donà

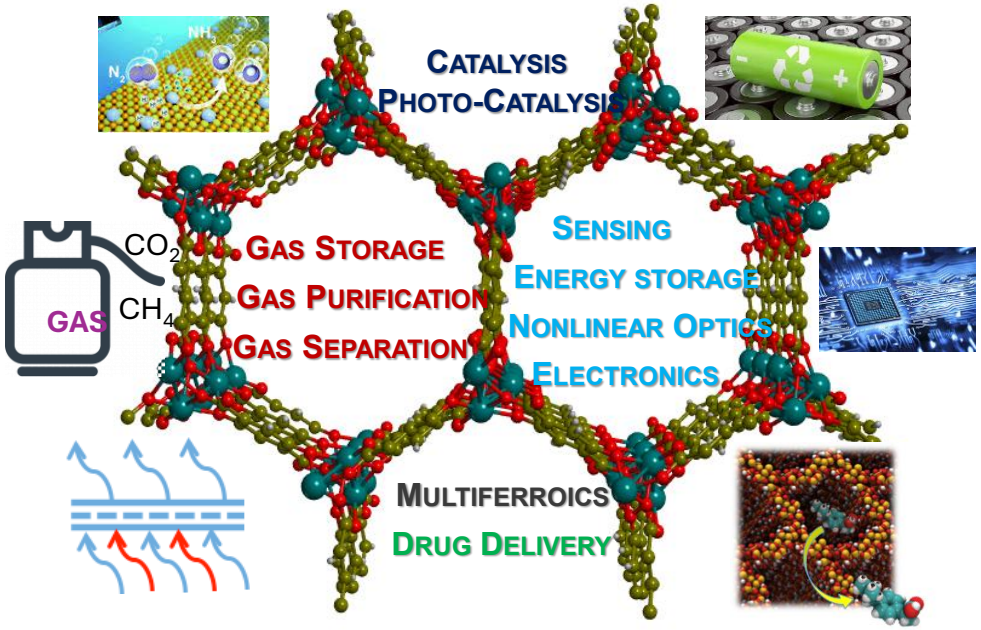


Lorenzo Donà, J. G. Brandenburg, I. J. Bush and B. Civalleri, Faraday Discuss., 224,292, **2020**

High-Throughput Screening: Metal-Organic Frameworks



- Inorganic Cluster
- Organic Linker
- 3D framework
- Porous material
- Crystalline structure
- Framework Modifications
- High chemical versatility and modular nature



Our accurate yet cost-effective composite methods make DFT calculations feasible for large and complex systems such as MOFs, enabling:

- **High-Throughput Screening**
- the generation of **extensive datasets** for Machine Learning applications

ML-derived atomic charges

H. Paja

1. Prediction of atomic charges
2. From big data to small data

A.S. Rosen et al. Matter, 4, 1578-1597, **2021**

ML-derived atomic charges

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1. **Prediction of atomic charges**
2. **From big data to small data**

QMOF database

- Experimentally derived MOFs
- 20 375 cleaned structures
- Structures with no more than 300 atoms in the primitive cell
- PBE-D3(BJ)/520eV

Structural descriptors from matminer

Chemical descriptors from mendeleev Mv

Random Forest Algorithm

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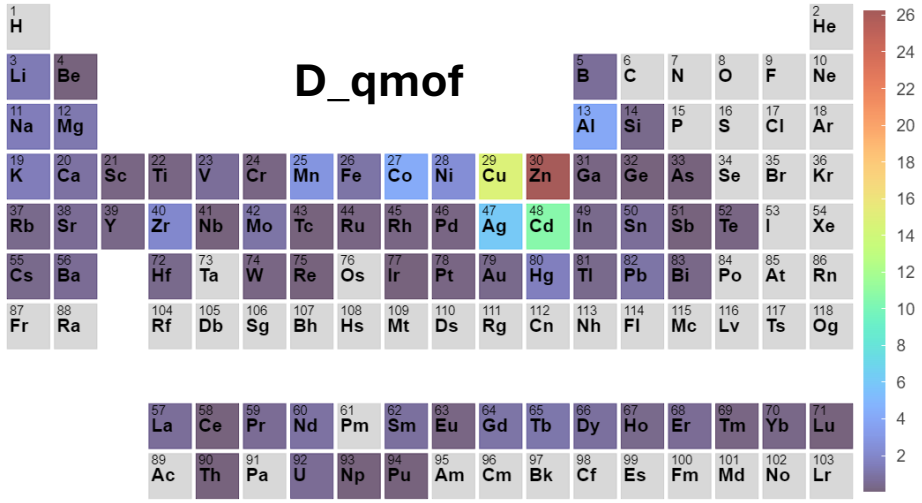

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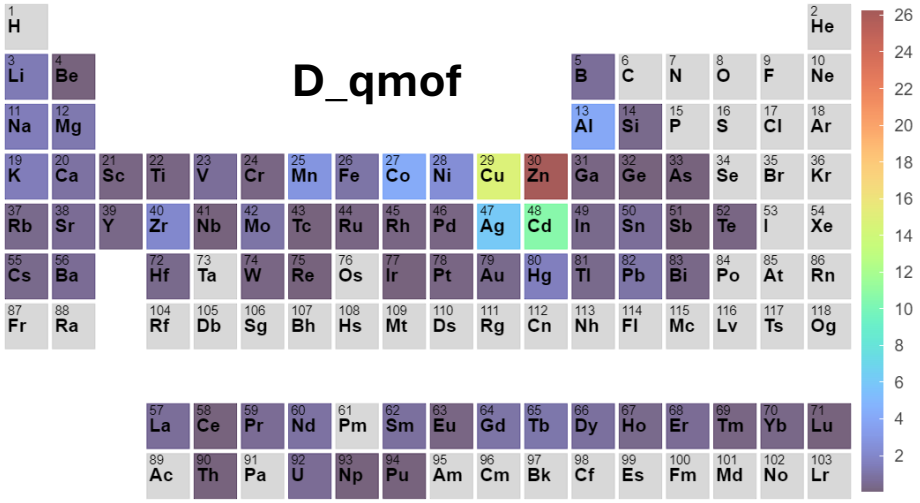
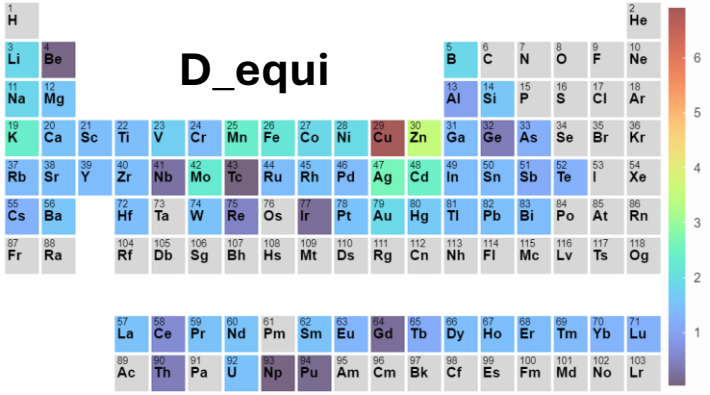


balanced composition with equated metal abundancy



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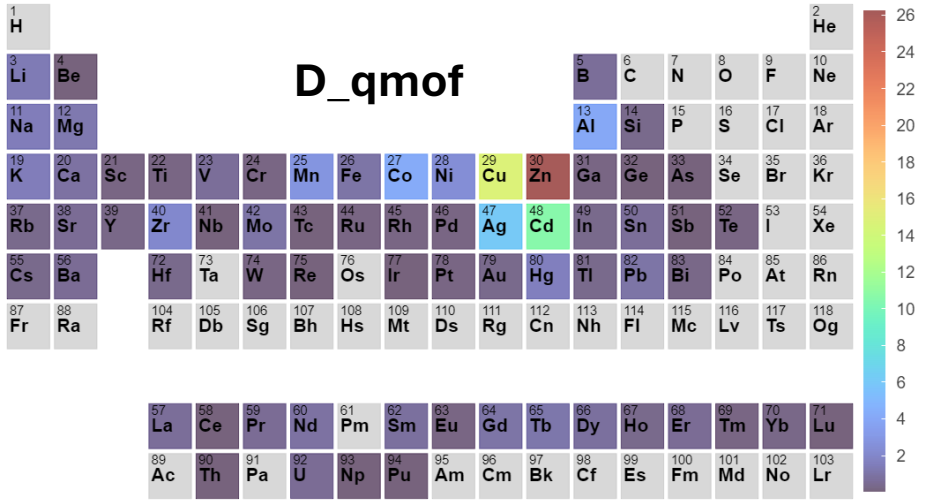

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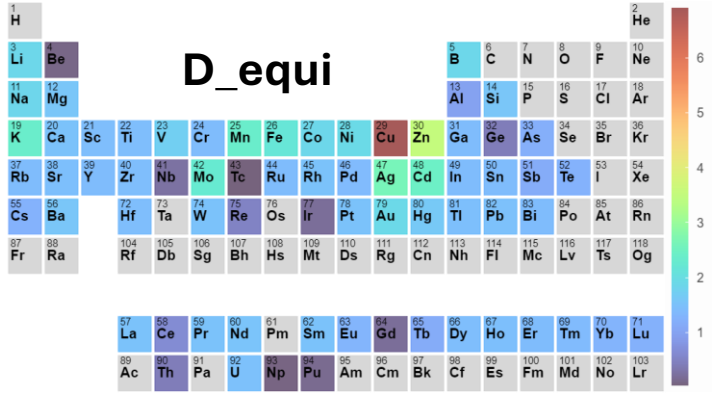
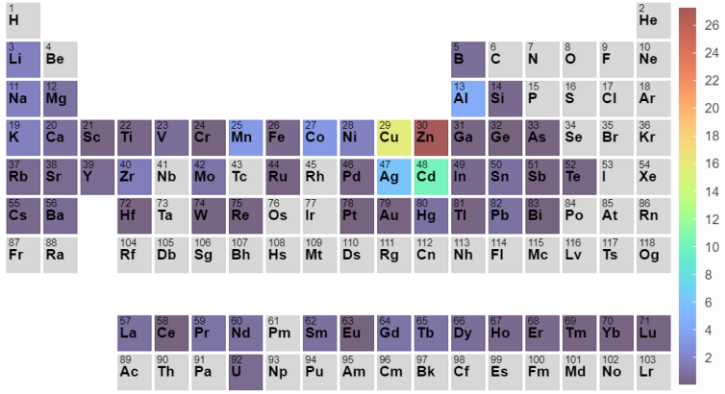


randomly selected spanning from 500 to 3000 MOFs



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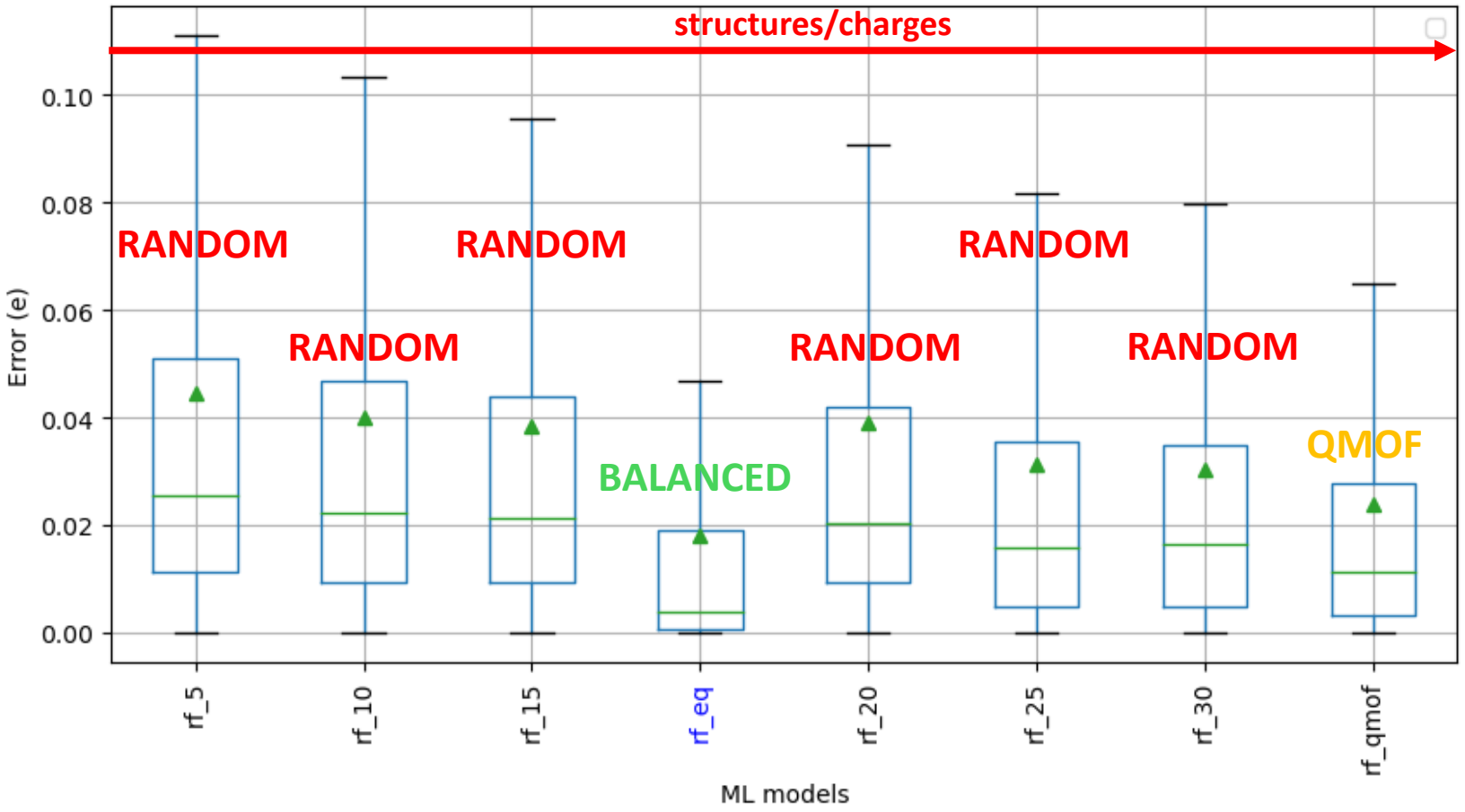

- Chemically-driven selection: more balanced composition with equated metal abundancy of 2%
- Although the size of the datasets increase, no relevant improvement on ML predicted charges are achieved.
- Despite this ***rf_eq*** represents a surprising improvement with a ***MAE*** lower than ***0.02 electrons.***

Label	5	10	15	EQ	20	30	QMOF
MOFs	500	1000	1500	1474	2000	3000	20309
DDEC	53 K	111 K	165 K	165 K	221 K	334 K	2 M

A.S. Rosen et al. Matter, 4, 1578-1597, **2021**

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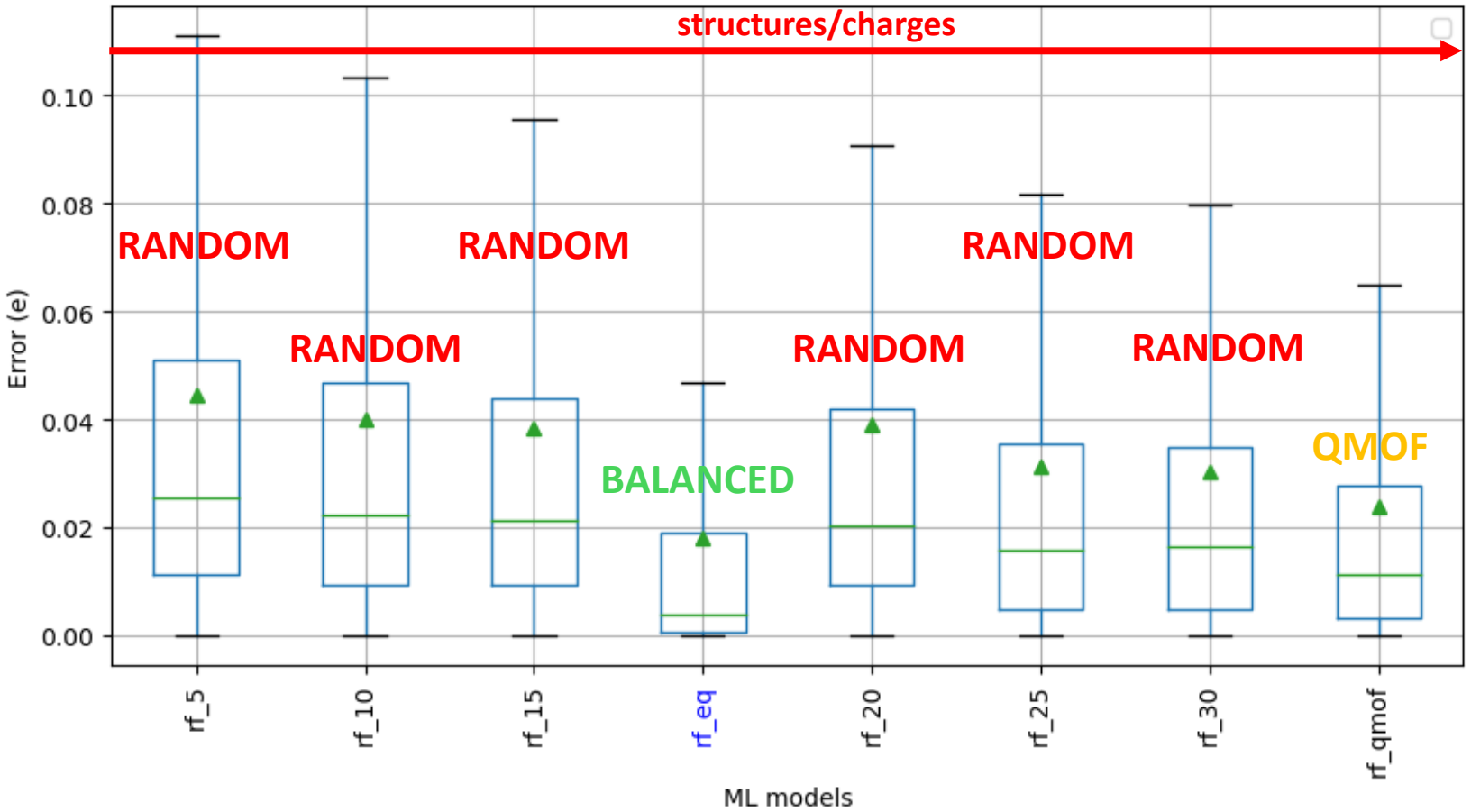

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For the balanced dataset we proceed to the calculation of **Hirshfeld-I** atomic charges using CRYSTAL23 with **r^{2}SCAN0pob-3c**, retrained the model, and achieved MAE of 0.06 electrons.

A.S. Rosen et al. Matter, 4, 1578-1597, **2021**

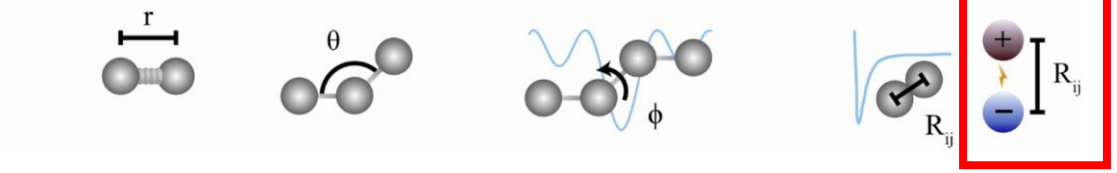
Adsorption Simulation

iRASPA
Molecular
Simulations

UFF force field

Bonded | Non-bonded

$$E_{total} = \sum_{bonds} K_r(r - r_{eq})^2 + \sum_{angles} K_\theta(\theta - \theta_{eq})^2 + \sum_{dihedrals} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{i<j} \left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^{6}} + \frac{q_i q_j}{\epsilon R_{ij}} \right]$$



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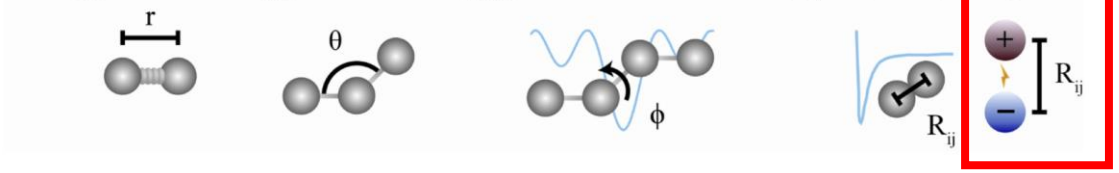
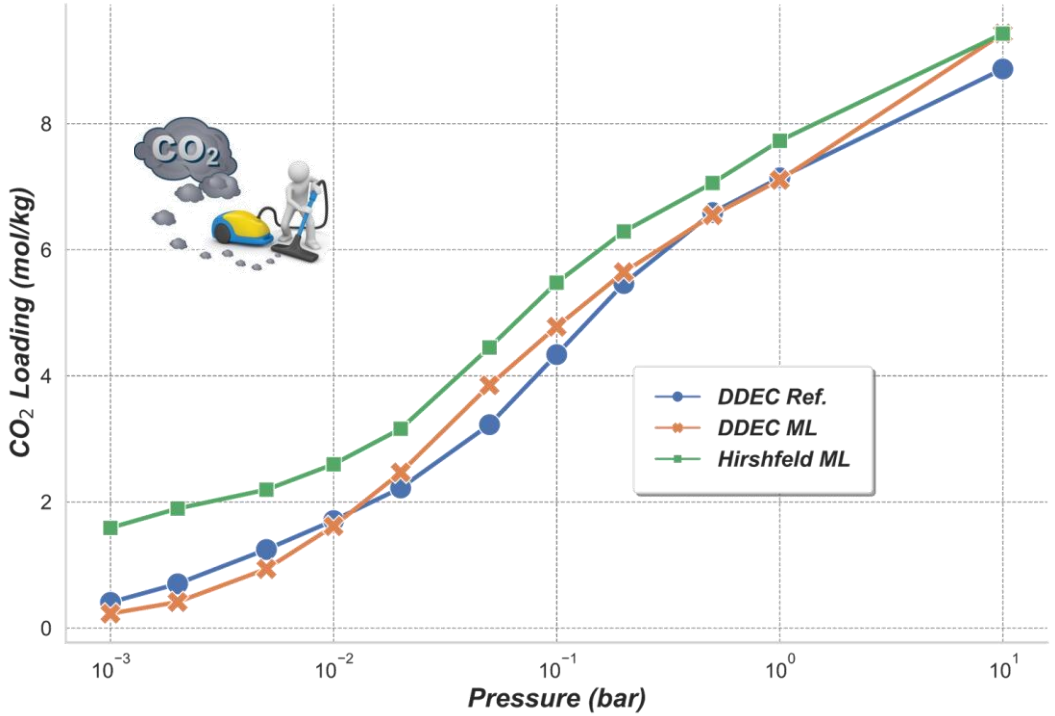
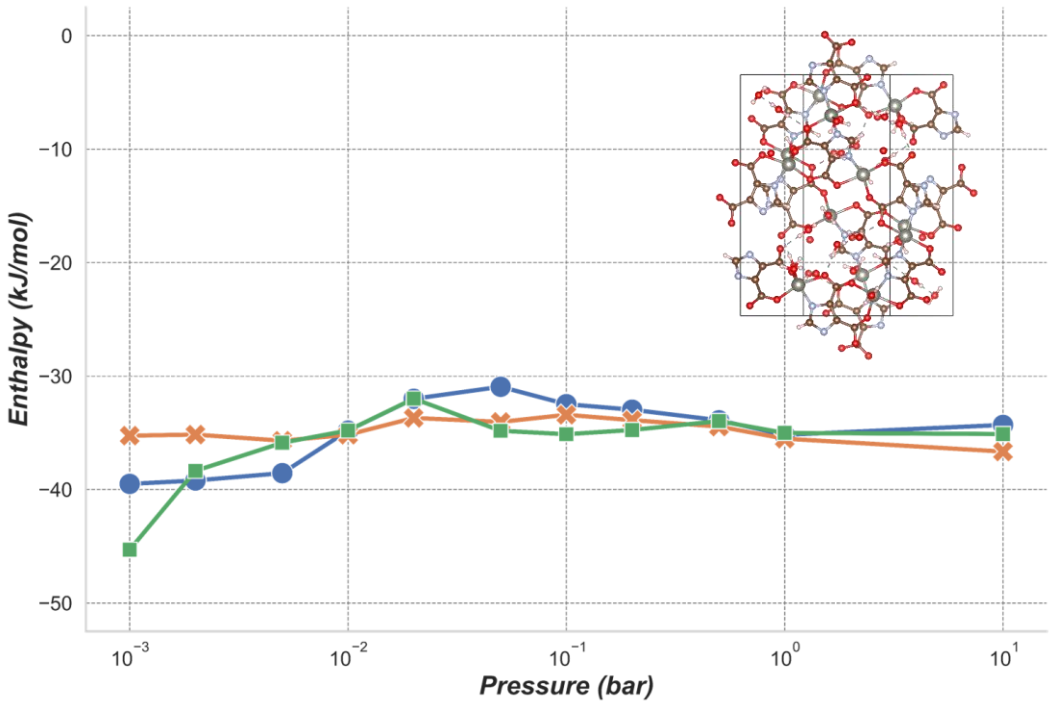


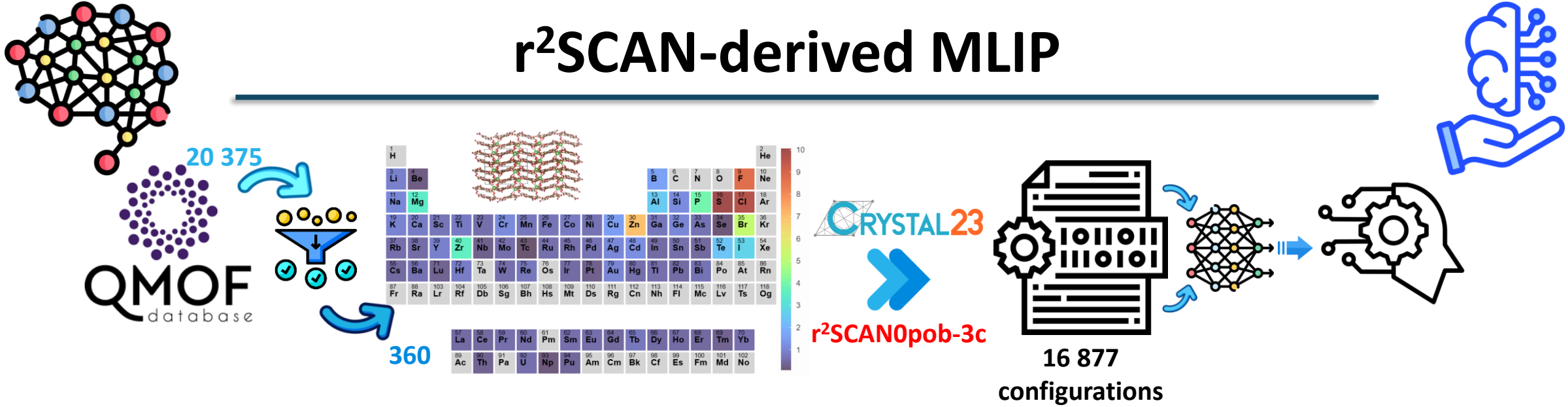




- Machine learning models can accurately predict Hirshfeld charges based on structural descriptors.
- These predicted charges can then be used to reliably perform GCMC simulations, producing results that are comparable to reference data.
- This approach can be scaled up to include a larger number of MOFs, improving our understanding of, and ability to design, materials for specific applications.

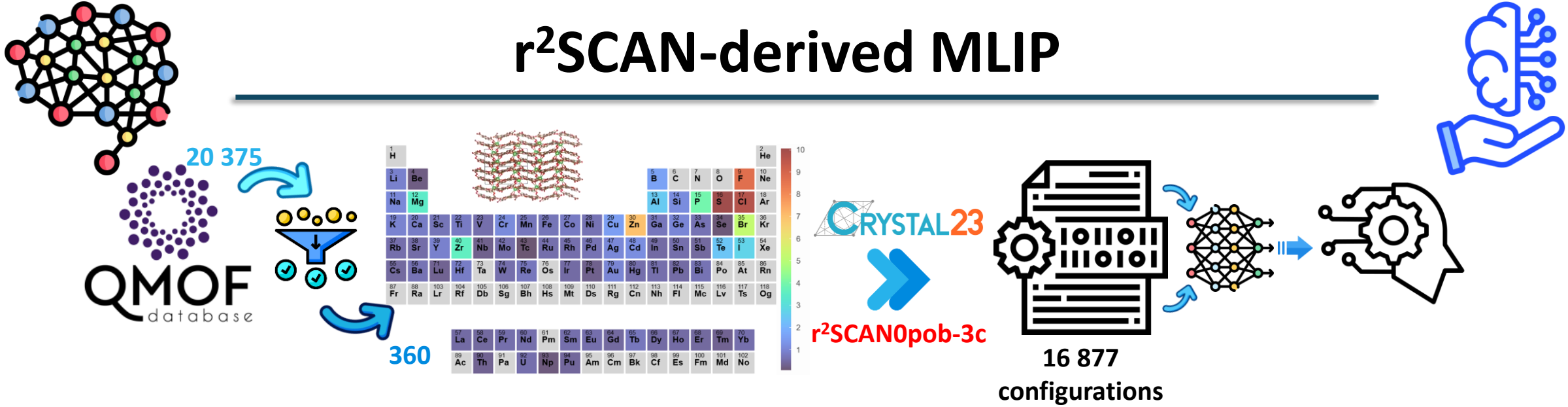
F. L. Oliveira et al., Scientific Data 10, **2023**

r²SCAN-derived MLIP



- ***r²SCAN0pob-3c*** method for Geometry Optimisation

r²SCAN-derived MLIP



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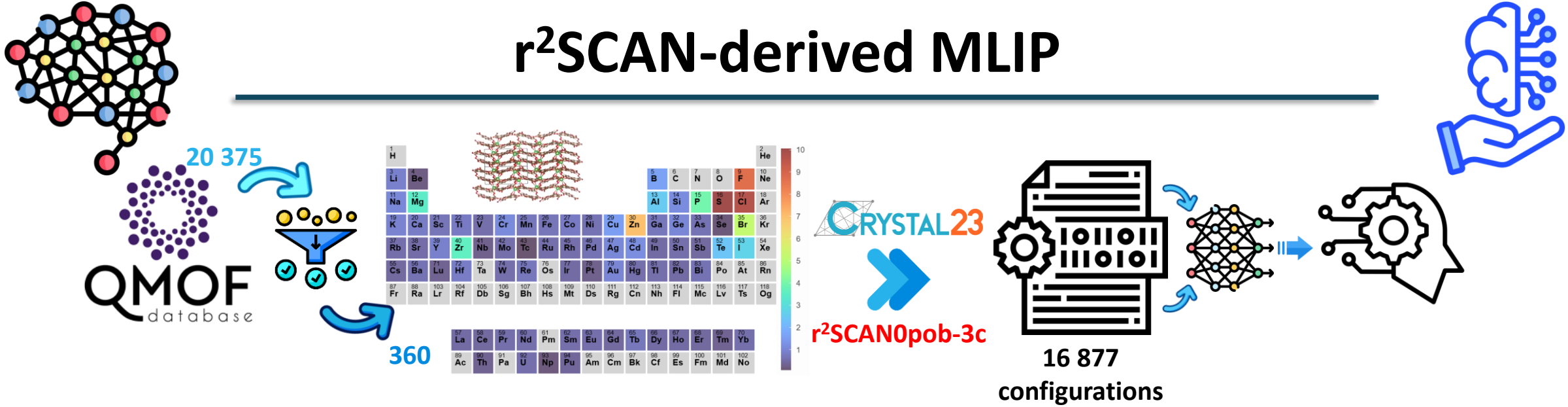

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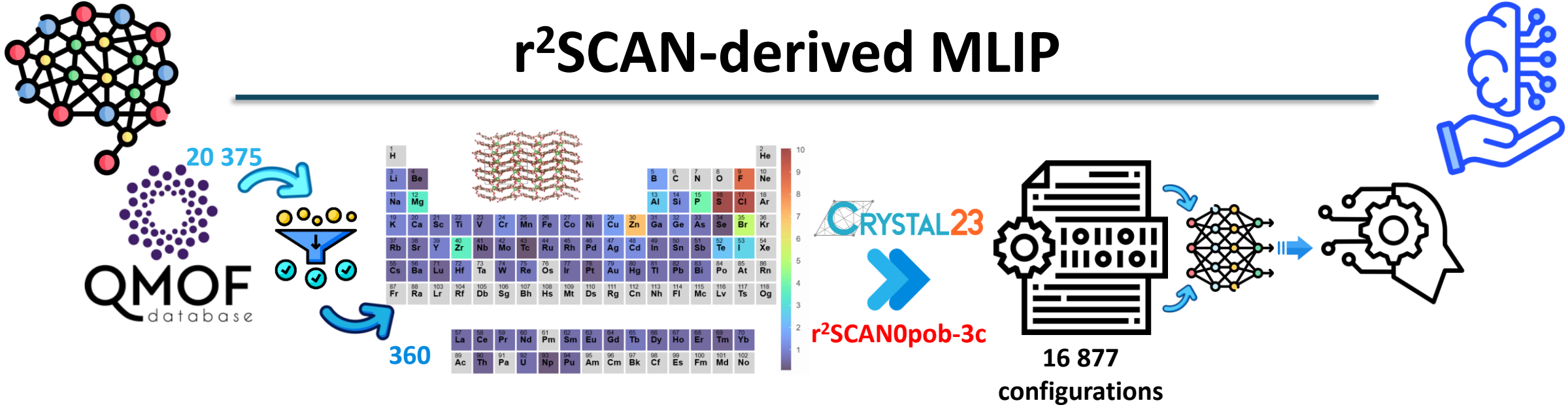


- ***r²SCAN0pob-3c*** method for Geometry Optimisation
- ***16877 MOF configurations*** with hundreds atoms in the primitive unit cell
- The training set comprises more than **5 million entries**, encompassing atomic positions, forces, and stress tensors.
- expanding the chemical design space with f-block elements
- general purpose universal **ML potential for MOFs**
- robustly validated

2 MACE model:

- **D8 (from scratch)**
- **D8_ft (finetuning of MP0)**

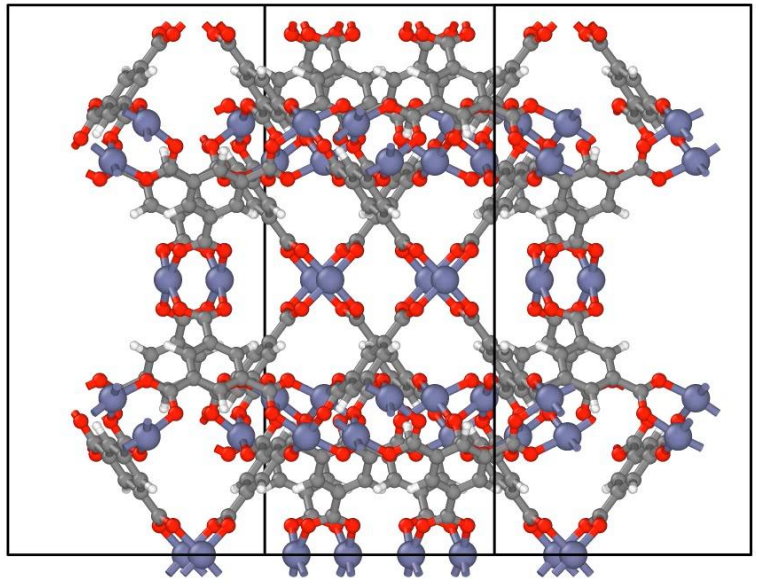
…pluggable with ASE, LAMMPS, Nequip, PyTorch…

Validation I

MOF-LC15 (***low Temperature X-ray diffraction measurements***)

Comparison with **MP0** and **OMAT** models

CPO-27-Mg	**CPO-27-Ni**	**CPO-27-Zn**
HKUST-1-Cu	**HKUST-1-Zn**	**MIL-125-Ti**
MIL-140-Zr	**MIL-47-V**	**MIL-53-Al-LP**
MIL-53-Al-NP	**MIL-68-Ga**	**MOF-5-Zn**
UIO-66-Zr	**UIO-66-Zr-OH**	**UIO-67-Zr-OH**



I.Batatia et al. https://doi.org/10.48550/arXiv.2206.07697 and https://doi.org/10.48550/arXiv.2401.00096.

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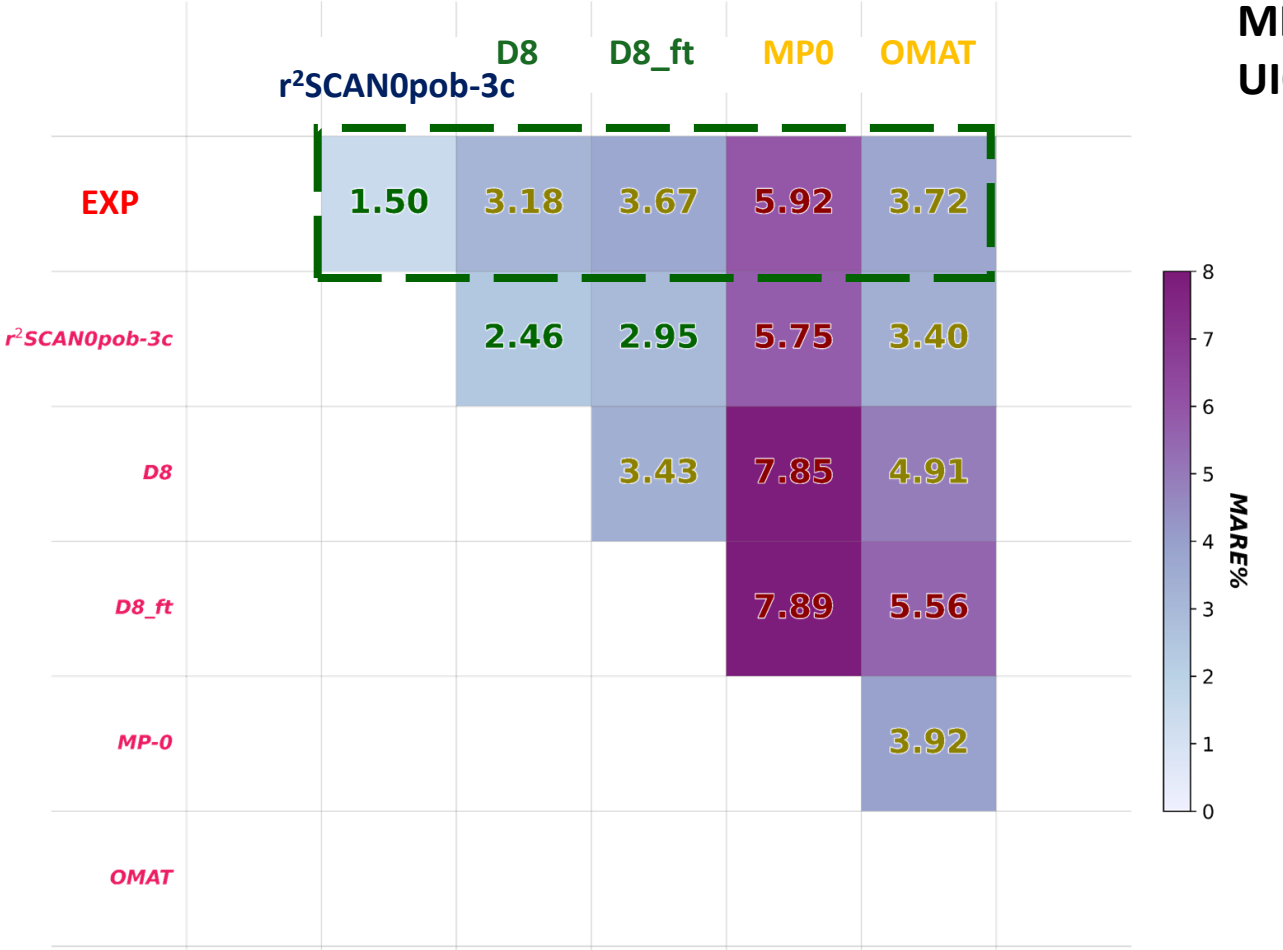

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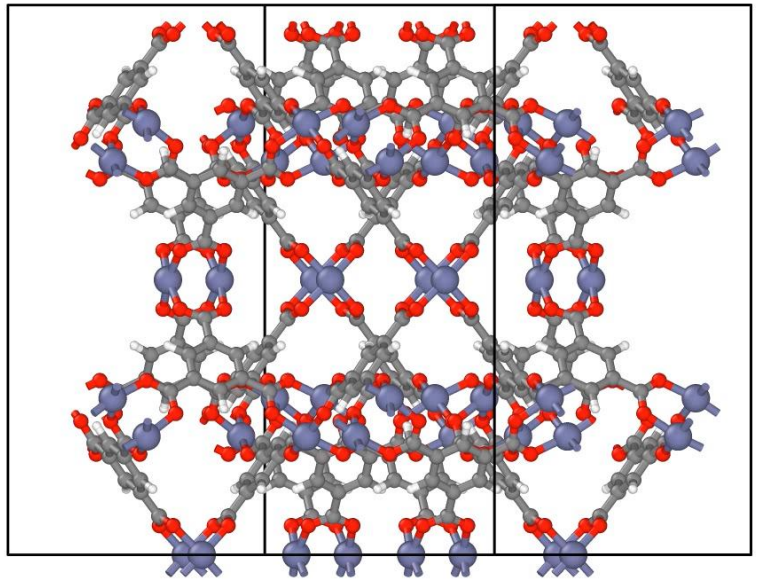
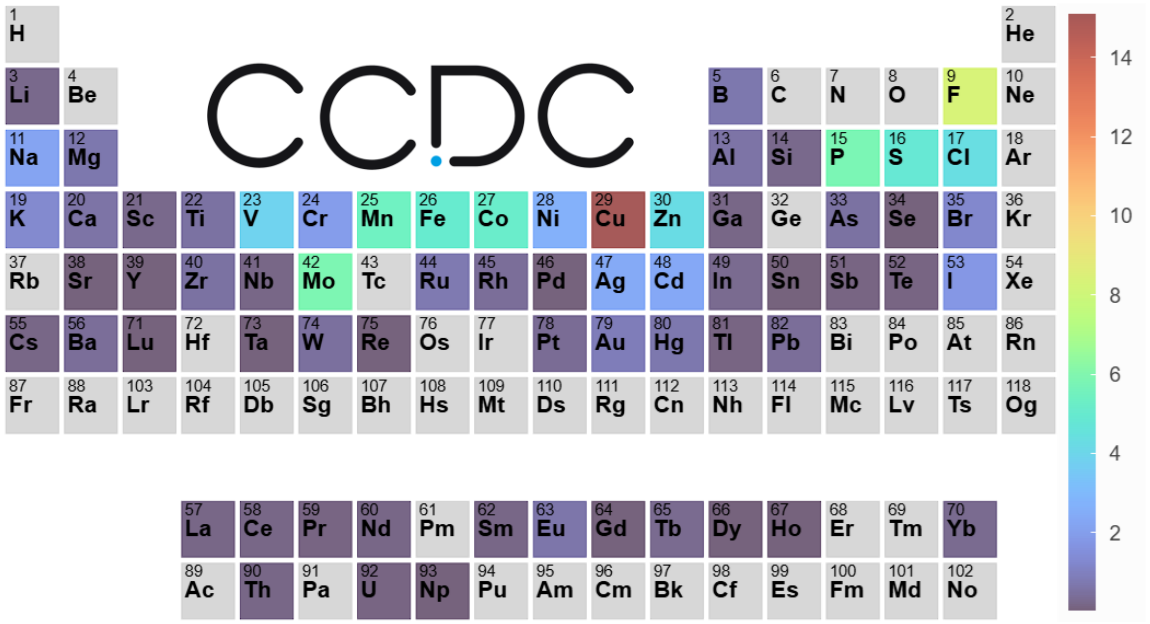
UIO-66-Zr	**UIO-66-Zr-OH**	**UIO-67-Zr-OH**

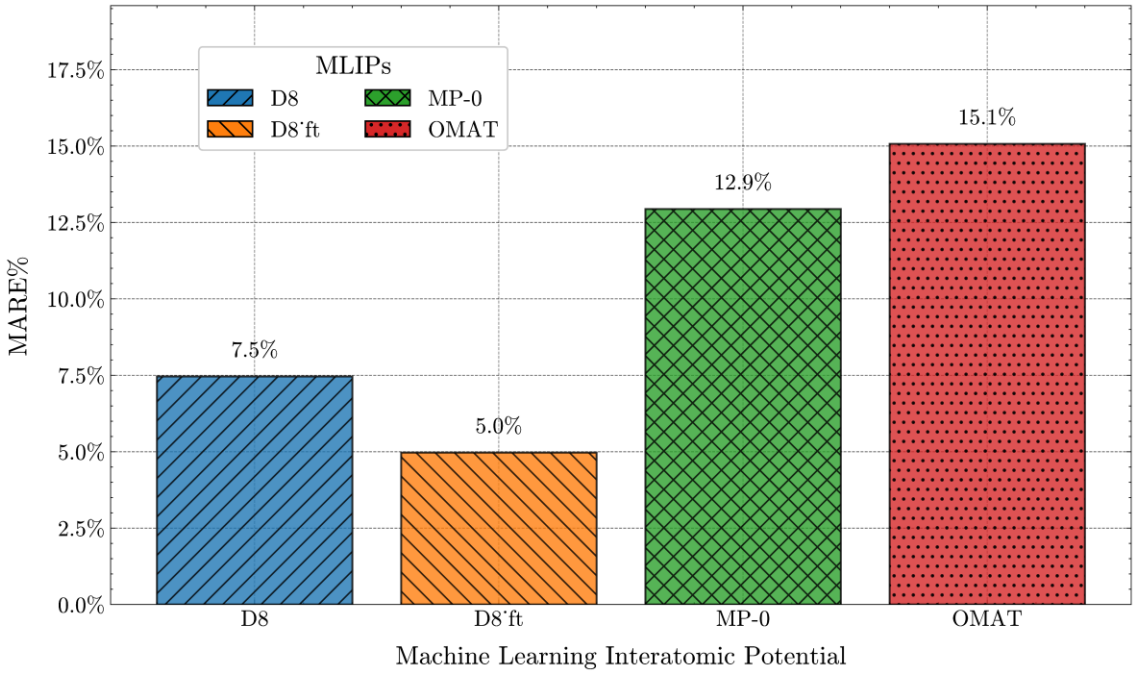


I.Batatia et al. https://doi.org/10.48550/arXiv.2206.07697 and https://doi.org/10.48550/arXiv.2401.00096.

Validation II

882 Experimental MOF from ***CCDC (T < 100 K and R-factors < 5%)***





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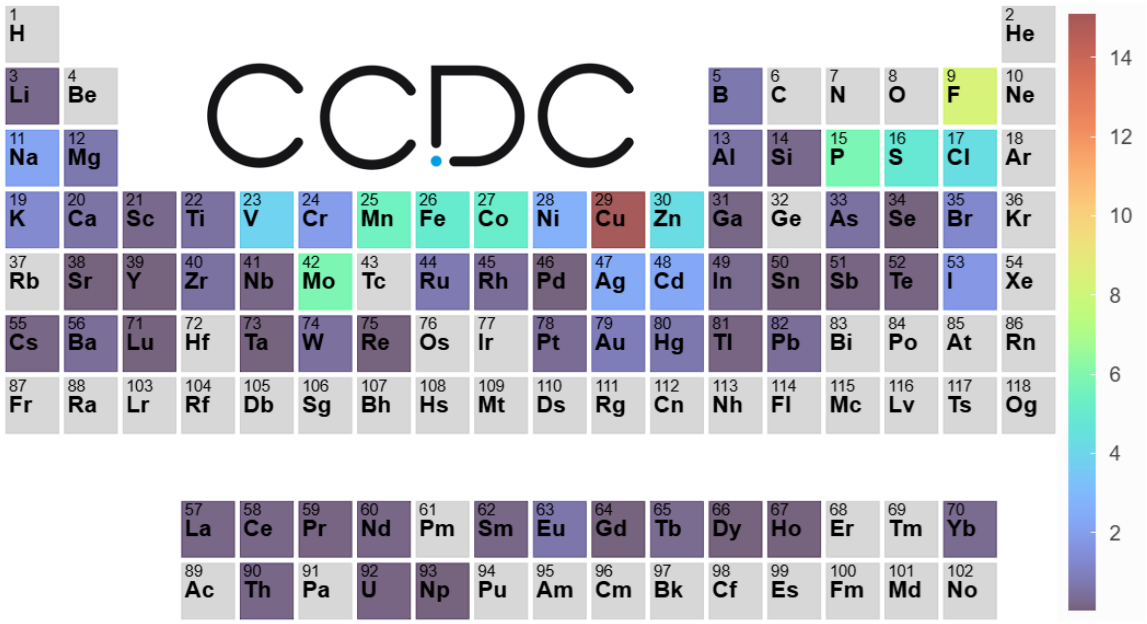

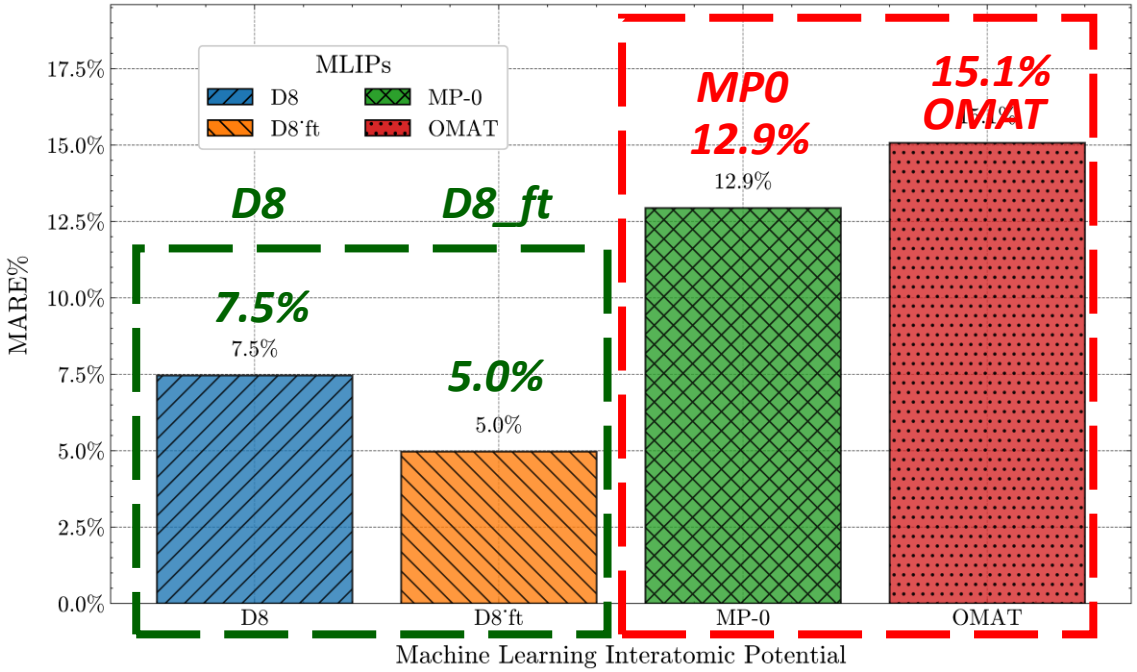

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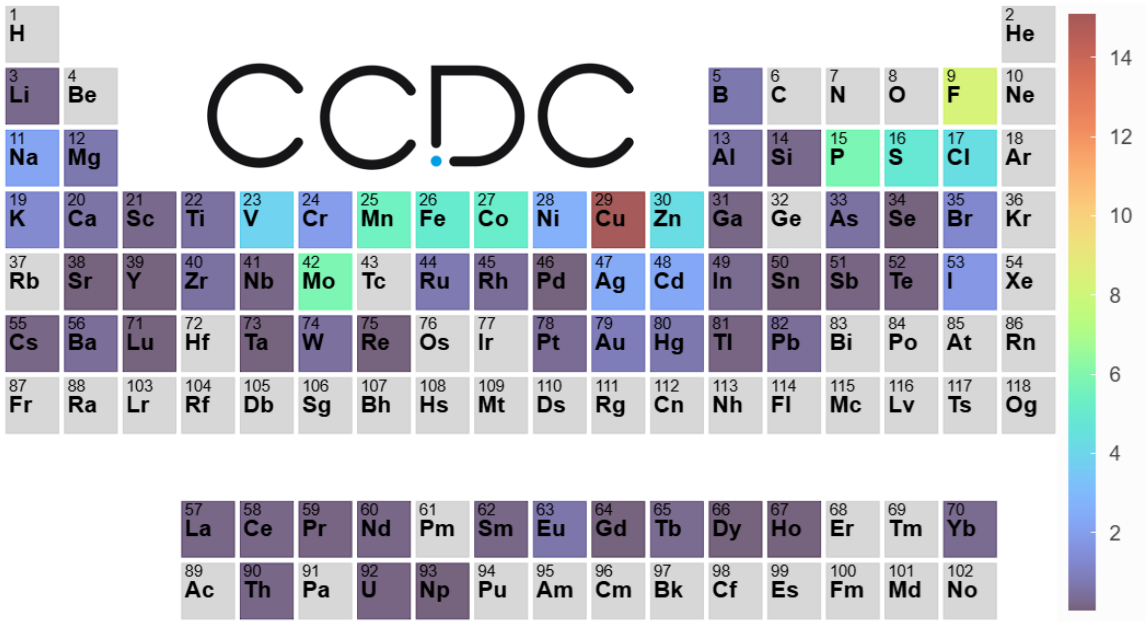
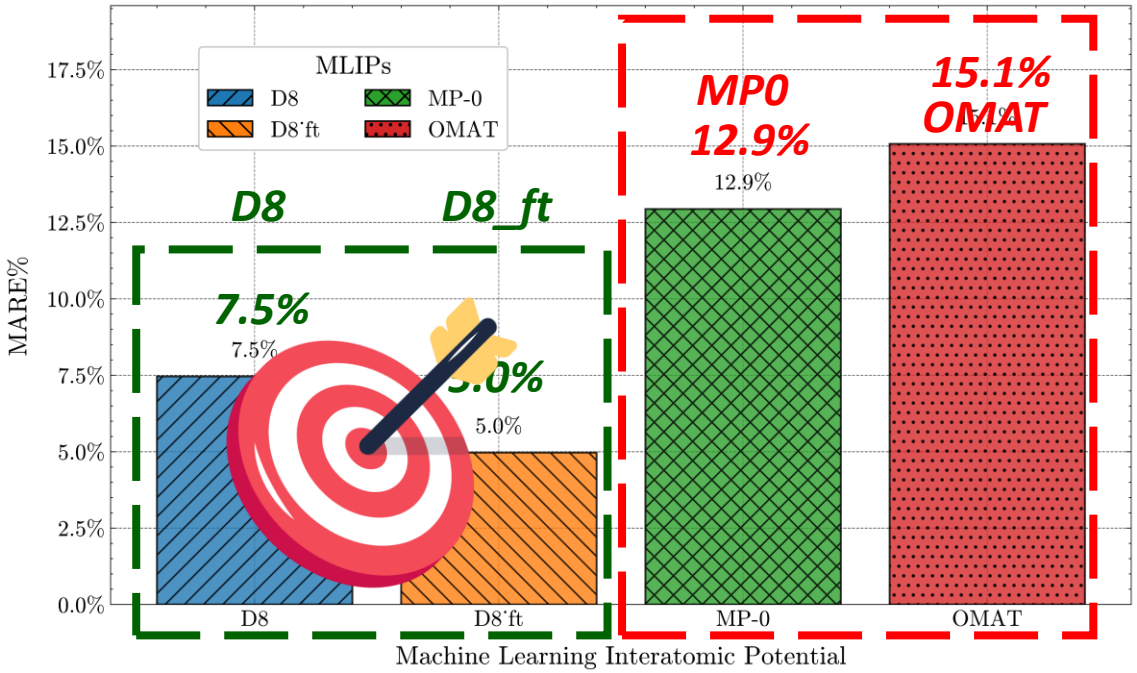


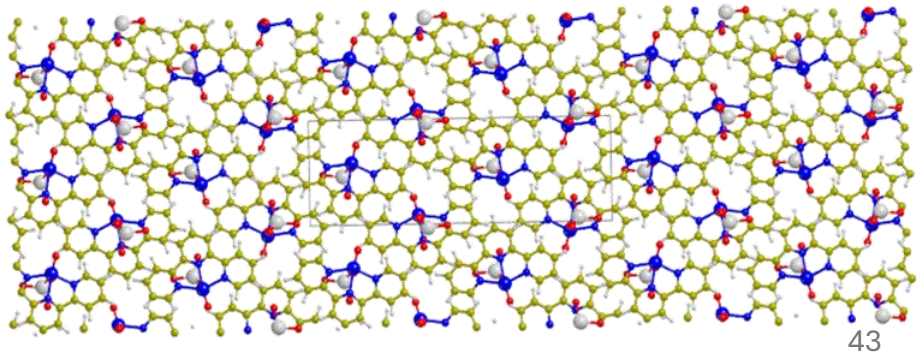


D8 model is about ***50 times faster*** than r^2SCAN0pob-3c

Optimisation time:

2x AMD EPYC 9534 64-Core Processor [256 Threads] | 100 GbE Ethernet Network | 768 GB RAM

UGARID: 144 atoms in the primitive cell



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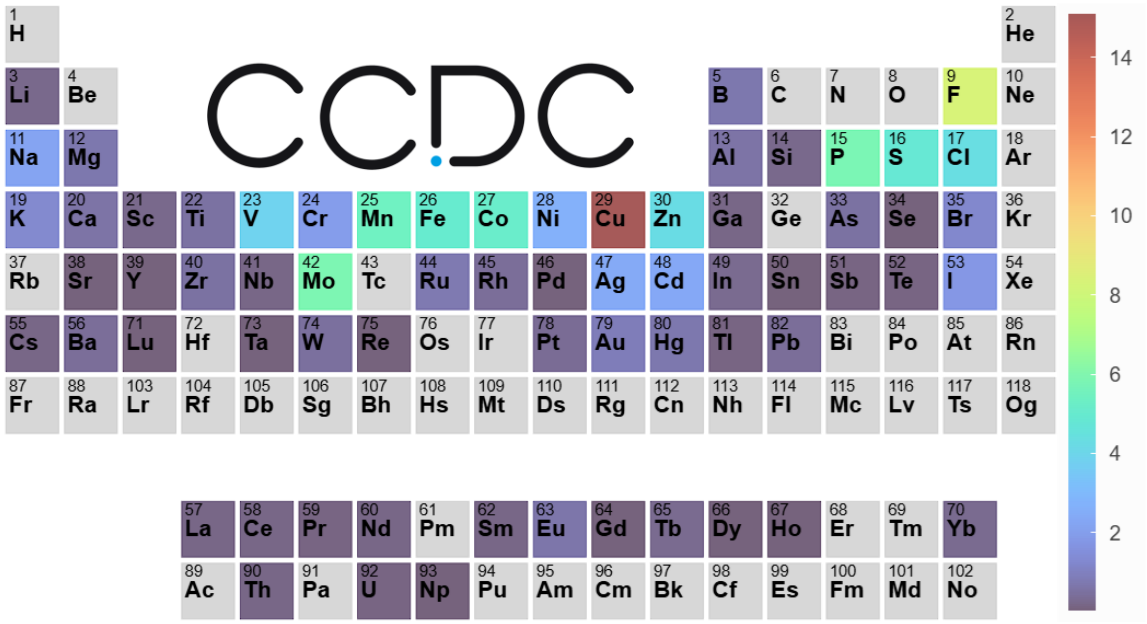
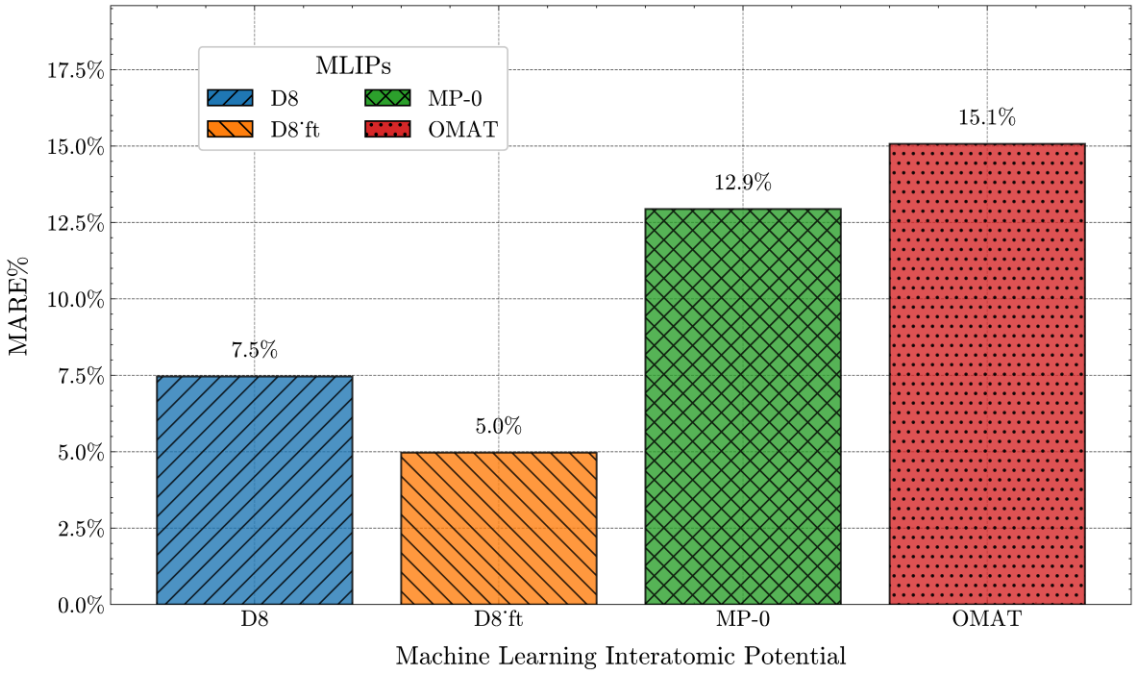




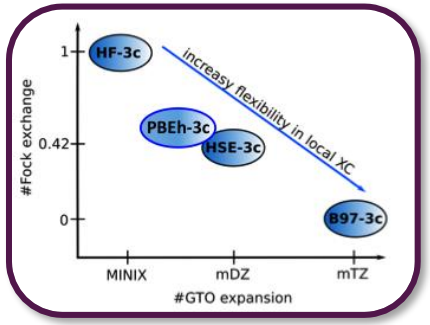
Despite their ***universal applicability***, both our MLIPs and state-of-the-art universals, such as MP-0 and OMAT, ***fail to optimise some MOFs***.

Future Perspective:

1. Expand Training Data
2. Active Learning Integration
3. Transfer Learning & Fine-Tuning
4. Hyperparameter Optimization

Acknowledgements

H. Paja L. Donà B. Civalleri



J. G. Brandenburg

I. Bush

D3(BJ) and gCP corrections

D3(BJ): London dispersion

$$E_{disp} = E^{(2)} + E^{(3)}$$

2-body damped atom-pair wise term

$$E_{disp}^{(2)} = -\frac{1}{2}\sum_{AB} \frac{C_6^{AB}}{R_{AB}^6 + f(R_{AB}^0)^6} + s_8 \frac{C_8^{AB}}{R_{AB}^8 + f(R_{AB}^0)^8}$$

Becke-Johnson damping function

$$f\left(R_{AB}^0\right) = a_1 R_{AB}^0 + a_2$$

3-body damped atom-triple wise term

$$E^{(3)} = \sum_{ABC} f_{d,(3)}(\bar{R}_{ABC}) E^{ABC}$$

gCP: geometrical counterpoise

$$E_{gCP} = \sigma \sum_{a}^{atoms} E_a^{atom}$$

Mimicks the Boys-Bernardi counterpoise method with an atom-pair wise term

$$E_{BSSE}^{gCP} = \sigma \sum_{A}^{atoms} \sum_{A \neq B}^{atoms} E_A^{miss} \frac{exp\left(-\alpha (R_{AB})^{\beta}\right)}{\sqrt{S_{AB} N_B^{virt}}}$$

E_A^{miss} is a gauge of the basis set incompleteness per each atomic species tuned through a decay function

$$f_{dec}\left(R_{AB,g}\right) = \frac{exp\left(-\alpha \left(R_{AB,g}\right)^{\beta}\right)}{\sqrt{S_{AB,g} N_B^{virt}}}$$